# Spatial Data Analysis Beyond Interpolation: A Functional Analytic Perspective on Geostatistics, Machine Learning and Geospatial Foundation Models

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## Abstract

This paper examines how geostatistics, machine learning, and geospatial foundation models can be combined for practical spatial prediction. It emphasizes common applied challenges: irregular sampling, spatial dependence, scale mismatch, support effects, and uncertainty. Rather than treating spatial analysis as simple interpolation, the chapter frames it as the task of estimating continuous environmental fields from incomplete observations and covariates. Geostatistical methods provide interpretable spatial dependence and uncertainty estimates, while machine learning captures nonlinear relationships in rich environmental data. A synthetic case study comparing Random Forest, Ordinary Kriging, and regression kriging shows that hybrid models can improve prediction by combining covariate-driven learning with residual spatial structure. I also discuss how geospatial foundation models trained on Earth Observation archives may support data fusion, multiscale mapping, and operational monitoring. The practical message is that robust spatial applications require models that jointly address prediction accuracy, spatial structure, scale, and uncertainty.

# Contents

# Chapter 1

# Introduction

Spatial phenomena are intrinsically continuous in space and time, a viewpoint that is central to geostatistical theory [31] and to practical spatial interpolation [23]. Observed datasets therefore represent finite and often irregular samples of an underlying spatial field, rather than independent observations drawn from an unstructured population [54].

Let

$$f : D \subset \mathbb{R}^d \to \mathbb{R} \tag{1.1}$$

denote a spatial process defined over a spatial domain $D$. The observed dataset

$$\{f(\mathbf{s}_1), \ldots, f(\mathbf{s}_n)\} \tag{1.2}$$

is only a finite sampling of this continuous object.

From this perspective, spatial modeling becomes an approximation problem defined over functional spaces, a mathematical setting formalized in functional analysis [110] and Hilbert-space methods [83]. The central question is not merely how to interpolate points, as in classical kriging [91], but how to infer a spatially structured function from incomplete and noisy observations [116].

Spatial datasets possess several properties that distinguish them from conventional tabular data. Nearby observations are often more similar than distant observations, a principle expressed by Tobler's law [121] and formalized through spatial autocorrelation [97, 26]. Directional effects may arise through anisotropy [54], sampling locations are often irregular in field surveys [130], and the support of observations can vary substantially across sensors, pixels, plots, and block averages [57].

These characteristics imply that spatial dependence is not a secondary technical detail. It fundamentally alters statistical inference [5], spatial prediction [31], model validation under spatial dependence [106], and uncertainty quantification [116]. Ignoring spatial structure can produce biased parameter estimates, misleading random validation scores [96], and maps with little statistical interpretability [54].

Historically, two major frameworks have dominated spatial prediction.

The first is geostatistics, which models spatial dependence explicitly through semivariogram theory [92] and covariance-based prediction [31]. The second consists of machine learning approaches, which emphasize flexible nonlinear prediction from explanatory variables [61] but often do not explicitly represent spatial covariance structure in the geostatistical sense [67].

These frameworks are often presented as competing paradigms. Mathematically, however, both can be interpreted as approximation procedures defined over functional spaces. Geostatistics emphasizes stochastic dependence, uncertainty propagation, and spatial covariance, whereas machine learning emphasizes nonlinear operator approximation in high-dimensional predictor spaces.

This chapter adopts a functional analytic perspective to establish a common mathematical foundation for both approaches. Spatial fields are interpreted as elements of infinite-dimensional spaces [110], kriging is interpreted as an orthogonal projection problem in Hilbert spaces [91], and covariance functions are interpreted as positive-definite kernels [6].

Under this interpretation, geostatistics and machine learning become closely related through reproducing kernel Hilbert spaces [6], regularization methods [129], kernel machine learning [113], and Gaussian process regression [104].

The remainder of the chapter is organized as follows. Section 2 introduces the functional analytic foundations of spatial modeling. Section 4 formalizes spatial dependence through covariance operators and stochastic fields. Subsequent sections compare geostatistical and machine learning approaches for spatial prediction, uncertainty quantification, multiscale analysis, and hybrid spatial modeling.

# Chapter 2

# Functional analytic foundations of spatial modeling

## 2.1 Spatial fields as elements of function spaces

Spatial analysis is often introduced through maps, interpolation, and covariance functions. At a deeper mathematical level, however, most spatial models can be formulated as problems defined over functional spaces, including Banach spaces in analysis [110] and Hilbert spaces in projection theory [83].

Let

$$D \subset \mathbb{R}^d \tag{2.1}$$

denote a spatial domain and let

$$f : D \to \mathbb{R} \tag{2.2}$$

represent a spatial field.

Rather than interpreting the observations merely as discrete samples,

$$\{f(\mathbf{s}_1), f(\mathbf{s}_2), \ldots, f(\mathbf{s}_n)\}, \tag{2.3}$$

functional analysis interprets the spatial process itself as an element of an infinite-dimensional function space, as in functional data analysis [102] and stochastic function-space formulations [69]. This viewpoint is natural in environmental sciences because variables such as temperature, precipitation, salinity, soil moisture, pollutant concentration, and biomass are intrinsically continuous. The observations are finite measurements of an underlying spatial continuum rather than isolated quantities.

Many spatial processes can therefore be represented in Banach spaces [13] such as

$$L^p(D), \tag{2.4}$$

equipped with norm:

$$\|f\|_p = \left( \int_D |f(\mathbf{s})|^p d\mathbf{s} \right)^{1/p}. \tag{2.5}$$

The choice of norm determines which properties of the field are emphasized. For example, $L^2$ spaces emphasize energy and variance [83], while Sobolev spaces additionally incorporate smoothness and differentiability constraints [1].

## 2.2 Banach and Hilbert spaces

A Banach space is a complete normed vector space [29]. Completeness ensures that convergent sequences remain inside the space, which is essential for the existence and stability of solutions in approximation problems [83] and inverse problems [39].

Among Banach spaces, Hilbert spaces play a central role in spatial statistics because they are equipped with an inner product:

$$\langle f, g \rangle = \int_D f(\mathbf{s}) g(\mathbf{s})\, d\mathbf{s}. \tag{2.6}$$

This induced geometry permits the definition of orthogonality and projection operators [83], spectral decompositions [29], and reproducing kernels [6]. Most classical geostatistical methods can be interpreted naturally inside Hilbert spaces of square-integrable random variables, especially when prediction is expressed through covariance-based linear estimation [31] and kriging systems [23].

## 2.3 Projection theory and spatial estimation

Let $\mathcal{H}$ denote a Hilbert space of random variables. Spatial prediction can then be formulated as an orthogonal projection problem.

Suppose observations are available at locations

$$\{\mathbf{s}_1, \dots, \mathbf{s}_n\}. \tag{2.7}$$

The predictor of the unobserved quantity

$$Z(\mathbf{s}_0) \tag{2.8}$$

is obtained by projection onto the subspace generated by the observations:

$$\mathcal{H}_n = \operatorname{span}\{Z(\mathbf{s}_1), \dots, Z(\mathbf{s}_n)\}. \tag{2.9}$$

The projection estimator is therefore

$$\hat{Z}(\mathbf{s}_0) = P_{\mathcal{H}_n} Z(\mathbf{s}_0), \tag{2.10}$$

where $P_{\mathcal{H}_n}$ denotes the orthogonal projection operator.

Under second-order assumptions, ordinary kriging emerges naturally as a linear unbiased spatial predictor [91] and as a minimum-variance projection estimator [31].

The kriging variance corresponds to the norm of the residual projection error:

$$\sigma_K^2 = \|Z(\mathbf{s}_0) - \hat{Z}(\mathbf{s}_0)\|_{\mathcal{H}}^2. \tag{2.11}$$

This interpretation is deeper than the traditional weighted-average formulation because it directly connects kriging with approximation theory [112], operator theory [83], and stochastic-process modeling [116].

## 2.4 Covariance kernels and reproducing kernel Hilbert spaces

Covariance functions are central objects in spatial modeling:

$$C(\mathbf{s}_i, \mathbf{s}_j) = \mathrm{Cov}\left[Z(\mathbf{s}_i), Z(\mathbf{s}_j)\right]. \tag{2.12}$$

From a functional analytic perspective, covariance functions are positive-definite kernels [6]. Such kernels induce reproducing kernel Hilbert spaces (RKHS) [6], which provide a common mathematical framework for smoothing and regularized estimation [129], kernel-based machine learning [113], and Gaussian process regression [104].

Under this interpretation, kriging, Gaussian process regression, support vector machines, and kernel ridge regression can all be viewed as kernel-based approximation procedures differing mainly in their assumptions concerning uncertainty, regularization, and optimization.

This equivalence is particularly clear in smoothing-spline theory, where penalty functionals define regularized estimators [129], and in Gaussian process regression, where covariance kernels determine posterior mean and uncertainty estimates [104].

This observation establishes an important theoretical bridge between geostatistics and machine learning.

## 2.5 Spatial inverse problems and regularization

Many spatial prediction problems are intrinsically ill-posed [39]. Sparse sampling [54], measurement noise [31], and scale mismatches [57] imply that multiple spatial fields may explain the observations equally well.

Let

$$A : \mathcal{H} \to \mathbb{R}^n \tag{2.13}$$

denote an observation operator and let

$$y = Af + \varepsilon \tag{2.14}$$

represent the observed data.

Recovering the unknown field $f$ from observations $y$ is generally an inverse problem in the deterministic regularization sense [39] and in the statistical/Bayesian inverse-problem sense [76].

A common regularized formulation is

$$\min_f \left[ \|Af - y\|^2 + \lambda \mathcal{R}(f) \right], \tag{2.15}$$

where:

- $A$ is the observation operator;
- $\mathcal{R}(f)$ is a regularization functional;
- and $\lambda$ controls smoothness or complexity.

Different regularization choices generate different classes of spatial models:

- Tikhonov regularization promotes smoothness [120];
- total variation regularization preserves discontinuities and sharp spatial boundaries [76];
- Sobolev regularization imposes differentiability constraints through Sobolev-space penalties [1].

These formulations are especially relevant for environmental sciences, where datasets are often incomplete [54], noisy [31], and multiscale [128].

## 2.6 Functional analysis and machine learning

Machine learning increasingly relies on functional analytic concepts. Neural networks can be interpreted as nonlinear function approximators [52], while deep learning optimization occurs in high-dimensional parameterized function spaces [86].

More recently, neural operators have extended this connection by learning mappings between functional spaces rather than only finite-dimensional vectors [88], while physics-informed neural networks incorporate differential-equation constraints directly into learning objectives [101].

This perspective is especially important in spatial-temporal modeling, where many environmental systems are governed by partial differential equations:

$$\mathcal{L}u = f, \tag{2.16}$$

where $\mathcal{L}$ is a differential operator.

Examples include groundwater flow, atmospheric transport, heat diffusion, contaminant propagation, and ocean circulation. The integration of geostatistics for stochastic spatial dependence [31], functional analysis for infinite-dimensional structure [110], inverse problems for reconstruction from indirect observations [39], and machine learning for nonlinear approximation [52] therefore represents a natural direction for next-generation spatial modeling.

# Chapter 3

# Geospatial Foundation Models and Learned Spatial Representations

## 3.1 From handcrafted features to learned representations

For several decades, spatial prediction from remote sensing data relied heavily on manually engineered features derived from physical intuition and domain expertise, including spectral vegetation indices [108, 122], soil and moisture indices [71, 47], and texture descriptors [60]. Rather than learning representations directly from raw observations, analysts constructed variables believed to capture relevant environmental processes, a workflow that remains common in applied remote sensing and spatial prediction [54, 130].

One of the most widely used examples is the Normalized Difference Vegetation Index (NDVI):

$$NDVI = \frac{\rho_{NIR} - \rho_R}{\rho_{NIR} + \rho_R}, \tag{3.1}$$

where $\rho_{NIR}$ and $\rho_R$ denote near-infrared and red reflectance, respectively. The index was introduced as a compact spectral proxy for vegetation condition and remains one of the most influential examples of handcrafted remote-sensing feature design [108, 122].

NDVI and related vegetation indices were designed to summarize complex spectral information into interpretable variables associated with vegetation vigor [122], biomass and canopy structure [108], and photosynthetic activity [71]. Similar approaches generated numerous spectral indices targeting specific environmental properties, including moisture content [47], vegetation condition [71], and water extent [94].

Beyond spectral indices, spatial analysis frequently incorporated texture measures derived from local neighborhoods, especially gray-level co-occurrence statistics [60] and multiscale wavelet representations [90]. Examples include variance filters for local heterogeneity, gray-level co-occurrence matrices for spatial texture [60], entropy measures for neighborhood complexity, and wavelet-based descriptors for multiscale structure [90]. These variables attempted to represent spatial organization, heterogeneity, and landscape structure using handcrafted transformations [60, 90].

From a machine learning perspective, these approaches can be interpreted as manually defining a feature map

$$\phi : X \to \mathbb{R}^p, \tag{3.2}$$

where $X$ denotes the original observation space and $\phi(X)$ represents engineered predictors.

Although these methods remain useful and physically interpretable, they possess an important limitation: the representation itself is fixed a priori, unlike representation learning approaches that estimate features from data [9]. The analyst determines which features are important before the learning process begins.

Recent developments in machine learning have shifted this paradigm through deep representation learning [86, 52]. Instead of manually constructing features, modern representation learning attempts to learn useful representations directly from data [9]. The objective is no longer merely to estimate a prediction function, as in standard supervised learning [61],

$$f(\mathbf{x}), \tag{3.3}$$

but also to learn an internal representation

$$\mathbf{z} = \Phi(\mathbf{x}), \tag{3.4}$$

where $\mathbf{z}$ captures latent structure relevant across many downstream tasks.

This transition mirrors developments in natural language processing, where large pretrained models learn reusable language representations [15], and computer vision, where vision transformers [38] and masked autoencoders [62] replaced many handcrafted visual descriptors. Earth observation is now undergoing a similar transformation [15, 38, 62].

## 3.2 Foundation models for Earth observation

The emergence of foundation models represents one of the most significant recent developments in Earth observation analytics, following the broader foundation-model paradigm described by **(author?)** [10]. Their appearance reflects a broader transition occurring across machine learning, where models are increasingly trained on massive and heterogeneous datasets before being adapted to specific downstream tasks, as in large language models [15] and visual pretraining systems [38, 62]. Rather than learning a single prediction objective, foundation models attempt to acquire general-purpose representations that capture regularities present across large collections of observations. The resulting representations can then be reused for many applications, reducing the need to design specialized models for each individual problem.

In the Earth observation domain, foundation models are typically trained using self-supervised objectives such as masked reconstruction [28] or large-scale geospatial pretraining on satellite imagery and related datasets [74]. The underlying idea is that the Earth system contains rich spatial, spectral, and temporal structure that can be learned directly from observations without requiring extensive manual labeling, paralleling self-supervised learning in vision [62] and representation learning more broadly [9]. By exploiting these large archives, the models learn internal representations that are expected to transfer across regions, sensors, environmental conditions, and analytical tasks.

Several prominent examples have emerged in recent years. AlphaEarth explores large-scale geospatial representation learning from heterogeneous Earth observation datasets, emphasizing the construction of representations that remain useful across multiple tasks and geographical contexts [14]. Prithvi follows a similar philosophy, leveraging large Earth observation archives to support applications such as land-cover mapping, environmental monitoring, disaster assessment, and climate-related analysis [74]. SatMAE extends masked autoencoder architectures to remote sensing imagery by reconstructing intentionally hidden portions of an image during training, thereby encouraging the model to learn meaningful spatial and spectral structure without relying heavily on labeled examples [28]. Clay likewise focuses on learning reusable geospatial embeddings from large collections of observations that can subsequently support a variety of downstream analyses [25].

Although these systems differ substantially in architecture, training strategy, and data sources, they share a common conceptual objective: transferable geospatial representation learning [10, 74]. Their primary goal is not to produce a specific environmental prediction, but rather to learn a representation space that captures useful information about the Earth system. Prediction, classification, segmentation, retrieval, anomaly detection, and change detection can then be performed on top of this learned representation. Consequently, the most important output of a foundation model is often not a map or a forecast, but a latent representation that can serve as a foundation for many subsequent tasks.

## 3.3 Embeddings as spatial operators

From a functional analytic perspective, foundation models can be interpreted as operators acting on spaces of Earth observations, paralleling neural-operator work on function-space mappings [82] and representation-learning views of neural networks [9]. Let

$$X \tag{3.5}$$

denote a space that may contain multispectral imagery, radar measurements, temporal sequences, topographic variables, or other forms of geospatial information. A foundation model defines a mapping

$$\Phi : X \to \mathbb{R}^d, \tag{3.6}$$

where the output dimension $d$ is typically much smaller than the dimensionality of the original observation space. The resulting vector

$$\mathbf{z} = \Phi(x) \tag{3.7}$$

is commonly referred to as an embedding and serves as a compact representation of the original observation.

Unlike traditional feature engineering, where the analyst explicitly specifies the transformation from observations to predictors, the mapping $\Phi$ is learned directly from data through representation learning [9] and deep neural optimization [52]. The embedding therefore emerges from the optimization process itself rather than from prior assumptions about which variables or indices should be important. In this sense, foundation models shift the focus from manually designing representations to learning them automatically from large collections of observations.

For geospatial systems, however, observations rarely exist independently of space and time, a central concern in spatial statistics [31] and spatiotemporal modeling [23]. The meaning of a measurement often depends not only on the observed signal but also on where and when it was acquired. A more realistic formulation is therefore

$$\Phi(x, \mathbf{s}, t), \tag{3.8}$$

where $x$ denotes the observed signal, $\mathbf{s}$ denotes spatial location, and $t$ denotes time. Under this interpretation, the embedding becomes a spatial-temporal representation that jointly incorporates information about the observation itself and its environmental context.

Such embeddings may implicitly encode spectral relationships [28], spatial patterns [74], temporal dynamics, land-surface structure, environmental similarity, and other latent properties that are difficult to describe through handcrafted features alone [9]. This perspective is important because it clarifies what foundation models actually learn. Their primary objective is not to estimate environmental variables directly, but to construct operators that transform observations into useful latent representations. Prediction then becomes a secondary task performed within the learned embedding space. Foundation models should therefore be understood principally as representation-learning systems rather than prediction systems.

## 3.4 Spatial inference remains an open problem

The rapid success of geospatial foundation models has generated considerable enthusiasm, especially around large-scale Earth observation pretraining [74, 14], and in some cases the perception that traditional spatial statistics may eventually become unnecessary. Such conclusions are difficult to justify. Foundation models unquestionably address an important component of the spatial inference problem by providing powerful mechanisms for extracting informative representations from large and heterogeneous datasets [10, 25]. However, representation learning [9] and spatial inference [31] are not identical problems, and many uncertainty challenges remain unresolved in modern machine learning [48].

One important example concerns spatial support, a classical change-of-support issue in spatial statistics [57]. An embedding extracted from a 10-meter satellite pixel, a 1-kilometer climate grid cell, or an areal average over an agricultural field does not automatically resolve the change-of-support problem. Even if the learned representations are highly informative, they may still correspond to fundamentally different observational supports. Consequently, many of the difficulties associated with comparing, fusing, or transferring information across scales remain present.

Scale dependence presents a closely related challenge, because spatial variance and covariance can change across resolutions [128]. Foundation models often appear capable of learning multiscale structure implicitly, yet this does not necessarily imply a formal understanding of how variance, covariance, or uncertainty evolve across scales. The model may successfully encode information from multiple resolutions without providing an explicit statistical framework describing the relationships among them. Similar considerations apply to spatial covariance, which geostatistics models explicitly through covariance and variogram functions [31, 54]. Two observations may occupy nearby locations in an embedding space while possessing very different covariance structures in the underlying physical field. Learned similarity and statistical dependence are related concepts, but they are not equivalent.

Perhaps the most important limitation concerns uncertainty, a central requirement for geostatistical prediction [116] and a known challenge in machine learning [48]. Most current foundation models produce deterministic embeddings of the form

$$\mathbf{z} = \Phi(x, \mathbf{s}, t), \tag{3.9}$$

which provide no direct characterization of prediction uncertainty, spatial uncertainty, support uncertainty, or model uncertainty. Yet these quantities remain central to scientific inference and environmental decision-making. A representation may be highly informative while still offering little guidance regarding confidence, reliability, or risk.

For this reason, foundation models should not be interpreted as replacements for geostatistics [31], uncertainty quantification [116], or spatial inference theory [54]. A more productive perspective is to view them as powerful new representation operators that can be integrated into broader spatial modeling frameworks. Within such a framework, foundation models provide learned representations [10], machine learning supplies flexible predictive mappings [61], and geostatistics contributes explicit treatment of spatial dependence and uncertainty [31]. The future of spatial analysis will likely emerge not from the dominance of any single paradigm, but from the integration of these complementary perspectives within a unified inferential framework.

# Chapter 4

# Spatial dependence and stochastic fields

## 4.1 Spatial random fields

Geostatistics models spatial phenomena as realizations of random fields, a foundation introduced in regionalized variable theory [92], developed in modern spatial statistics [31], and widely used in applied geostatistical modeling [23].

Let

$$Z(\mathbf{s}), \quad \mathbf{s} \in D, \tag{4.1}$$

denote a spatial random field.

A common decomposition is

$$Z(\mathbf{s}) = m(\mathbf{s}) + \varepsilon(\mathbf{s}), \tag{4.2}$$

where:

- $m(\mathbf{s})$ represents the deterministic trend;
- $\varepsilon(\mathbf{s})$ denotes a zero-mean spatially correlated residual process.

This decomposition separates large-scale deterministic variation, often represented as a trend or drift [92], from local stochastic dependence represented through residual spatial covariance [54]. Similar mean-plus-residual decompositions are also central in model-based geostatistics [37].

## 4.2 Spatial autocorrelation

The defining property of spatial data is spatial autocorrelation: nearby observations tend to be more similar than distant observations, a pattern measured by Moran's coefficient [97] and formalized in spatial statistical analysis [26]. This principle is often summarized through Tobler's first law of geography [121]:

> "Everything is related to everything else, but near things are more related than distant things."

Spatial autocorrelation implies that observations cannot generally be treated as independent samples in spatial econometrics [5] or geostatistical inference [31]. Ignoring this dependence can lead to:

- **Biased uncertainty estimates:** when nearby observations contain partly redundant information, treating them as independent can make confidence intervals and prediction intervals appear too narrow, especially when spatial covariance is ignored [116].
- **Inflated validation scores:** random train–test splits may place neighboring observations in both sets, allowing the model to benefit from spatial proximity rather than demonstrating true predictive transfer to new areas [106].
- **Unstable parameter inference:** regression coefficients, covariance parameters, and variable-importance measures can change substantially when spatial clustering [5] or uneven sampling density [130] is not accounted for.
- **Misleading predictive maps:** maps may look visually smooth and plausible while hiding spatially structured errors, local bias, or unsupported extrapolation beyond the sampled region [54].

## 4.3 Covariance and semivariogram functions

The covariance structure of a spatial field, central to geostatistical prediction [31] and Gaussian random field theory [116], is represented by

$$C(\mathbf{h}) = \mathrm{Cov}\left[Z(\mathbf{s}), Z(\mathbf{s}+\mathbf{h})\right]. \tag{4.3}$$

Under intrinsic stationarity, spatial dependence is equivalently represented through the semivariogram, introduced in early kriging theory [91] and formalized in regionalized variable theory [92]:

$$\gamma(\mathbf{h}) = \frac{1}{2}\mathrm{Var}\left[Z(\mathbf{s}) - Z(\mathbf{s}+\mathbf{h})\right]. \tag{4.4}$$

The semivariogram quantifies how dissimilarity increases as the separation vector $\mathbf{h}$ between two locations becomes larger [54]. It is one of the main diagnostic tools used to summarize spatial continuity in applied soil and environmental mapping [130] and to choose an appropriate covariance or variogram model [23].

Three quantities are commonly interpreted:

- the **nugget**, which represents apparent variation at very short distances and is commonly associated with measurement error [31], sampling error [130], or microscale variability below the sampling resolution [54];

- the **sill**, which is the plateau approached by the semivariogram and approximates the total variance of a stationary spatial process [73];

- the **range**, which is the distance at which the semivariogram reaches the sill and beyond which spatial dependence is weak or negligible [54].

For many stationary spatial processes [31], semivariance increases with distance and then stabilizes at a constant value, commonly denoted $C_0 + C$, where $C_0$ is the nugget and $C$ is the structured spatial variance [73]. The distance at which this plateau is reached is the range of spatial dependence, $R$ [54]. Pairs of observations separated by more than this range are usually treated as effectively uncorrelated for modeling purposes because their dissimilarity is no longer explained by spatial proximity [23]. When the semivariogram reaches a clear sill, the process can often be modeled as second-order stationary over the study domain [31]. By contrast, a semivariogram that continues to increase without reaching a plateau may indicate trend [54], non-stationarity [111], or a spatial scale larger than the sampling domain [130].

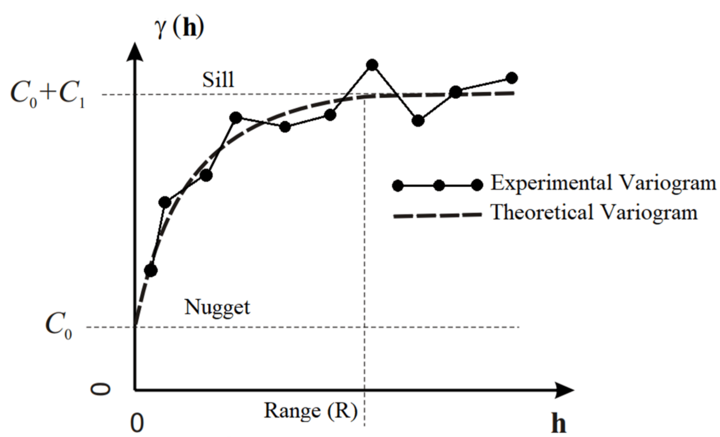


**Figure 4.1. Idealized semivariogram components.** The nugget marks discontinuity near the origin, the sill represents the variance plateau, and the range identifies the distance beyond which spatial dependence becomes weak.

Under ideal conditions, two measurements taken at exactly the same location would have zero separation distance and zero semivariance, so the semivariogram would start at the origin $(0, 0)$. In practice, experimental semivariograms often show a positive value as $h$ tends to zero. This discontinuity is called the nugget effect. It represents variation that cannot be resolved at the available sampling distance, including microvariability occurring below the shortest sampling lag, measurement error, positioning error, or other random sampling effects.

Variogram analysis remains one of the central tools of geostatistics because it provides direct information about scale-dependent spatial organization in applied spatial surveys [130] and in theoretical geostatistical modeling [23].

## 4.4 Stationarity and anisotropy

Most classical geostatistical methods rely on stationarity assumptions, especially second-order stationarity for covariance modeling [31]. Second-order stationarity requires:

$$\mathbb{E}[Z(\mathbf{s})] = \mu, \tag{4.5}$$

and

$$\mathrm{Cov}\left[Z(\mathbf{s}), Z(\mathbf{s}+\mathbf{h})\right] = C(\mathbf{h}), \tag{4.6}$$

meaning that covariance depends only on spatial separation. These assumptions provide the mathematical conditions under which covariance models can be estimated [31] and variogram models can be used for spatial prediction [23]. In practice, many environmental processes violate strict stationarity because of trends [111], discontinuities, or heterogeneous physical controls [8]. Anisotropy introduces additional complexity because spatial dependence may vary by direction rather than distance alone [54], requiring direction-sensitive variogram modeling [23]. These properties are not merely diagnostic details. They determine whether a spatial covariance model is mathematically defensible.

## 4.5 Scale dependence and multiscale variability

Spatial dependence is inherently multiscale, a key motivation for nested variogram models and factorial kriging [128]. Environmental systems often exhibit:

- **Short-range local variability:** fine-scale fluctuations may arise from measurement noise [31], microclimate, soil heterogeneity [130], vegetation patches, or other processes that change over very small distances.

- **Intermediate-scale structures:** neighborhoods, watersheds, landforms, management units, or ecological zones can create coherent spatial patterns that are larger than point-level noise but smaller than broad regional trends [128].

- **Large-scale deterministic gradients:** regional changes in elevation, climate, latitude, geology, or land-use intensity can generate smooth spatial trends that dominate the mean structure of the field [54].

A single covariance structure may therefore fail to represent all relevant spatial scales adequately in nested spatial processes [128] or in covariance-based prediction [31]. This motivates multiscale approaches such as nested variograms [128], factorial kriging [53], and multiscale covariance constructions [51]. Understanding scale dependence is fundamental because predictive performance [54], uncertainty quantification [116], and interpretability [130] all depend strongly on the scale at which spatial organization is modeled.

# Chapter 5

# Geostatistical modeling

## 5.1 From covariance structure to spatial prediction

Geostatistics uses the covariance structure of spatial fields [31] to construct optimal linear predictors at unsampled locations, following the kriging framework introduced in early kriging theory [91] and developed in regionalized variable theory [92].

Let observations be available at irregular sampling locations [130]:

$$\{\mathbf{s}_1, \ldots, \mathbf{s}_n\}, \tag{5.1}$$

and let

$$Z(\mathbf{s}_0) \tag{5.2}$$

denote the unknown value at target location $\mathbf{s}_0$, the standard point-prediction target in kriging [54].

Ordinary kriging estimates the target value as a weighted linear combination of neighboring observations [73]:

$$\hat{Z}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i), \tag{5.3}$$

where the weights $\lambda_i$ are determined by minimizing the prediction variance [31] subject to unbiasedness constraints [23].

The kriging system is obtained from the covariance structure of the field [116] and yields the best linear unbiased estimator (BLUE) under second-order assumptions [31]. Unlike deterministic interpolation procedures [73], kriging explicitly quantifies prediction uncertainty through the kriging variance [54]:

$$\sigma_K^2(\mathbf{s}_0) = \mathrm{Var}\left[ Z(\mathbf{s}_0) - \hat{Z}(\mathbf{s}_0) \right]. \tag{5.4}$$

This uncertainty estimate is fundamental because two predictions with identical expected values may possess very different confidence levels depending on local sampling density [130] and covariance structure [31].

## 5.2 Ordinary, universal, and regression kriging

Different kriging variants correspond to different assumptions concerning the mean structure of the spatial field [54].

Ordinary kriging assumes an unknown but locally constant mean [92]:

$$Z(\mathbf{s}) = \mu + \varepsilon(\mathbf{s}). \tag{5.5}$$

Universal kriging incorporates deterministic spatial trends or drift terms [23]:

$$Z(\mathbf{s}) = m(\mathbf{s}) + \varepsilon(\mathbf{s}), \tag{5.6}$$

where the drift is represented by basis functions or spatial covariates [31]:

$$m(\mathbf{s}) = \sum_{k=1}^{p} \beta_k f_k(\mathbf{s}). \tag{5.7}$$

Regression kriging combines regression models [68] with geostatistical modeling of residual spatial dependence [63]:

$$Z(\mathbf{s}) = f(\mathbf{x}(\mathbf{s})) + \varepsilon(\mathbf{s}), \tag{5.8}$$

where:

- $f(\mathbf{x}(\mathbf{s}))$ captures deterministic covariate effects [67];
- $\varepsilon(\mathbf{s})$ captures remaining spatial autocorrelation [31].

This formulation already suggests a bridge between geostatistics and machine learning through residual kriging [64] and spatial prediction with covariates [65].

## 5.3 Sequential Gaussian simulation

Kriging produces smooth optimal estimates [54] but tends to underestimate local variability because predictions correspond to conditional means [36]. In many environmental and geological applications, reproducing the full spatial variability [75] is more important than obtaining a single minimum-variance estimate. Sequential Gaussian Simulation (SGS) addresses this problem by generating conditional realizations of the spatial field [35]. The method assumes that the transformed variable follows a Gaussian distribution after normal-score transformation [23]:

$$Y(\mathbf{s}) = \phi\left(Z(\mathbf{s})\right), \tag{5.9}$$

where $\phi(\cdot)$ denotes the Gaussian anamorphosis transform [105] used to map the original variable to a Gaussian scale [54].

Simulation then proceeds sequentially over the domain [36]:

1. a random path through all unsampled locations is generated [35];
2. simple kriging is performed at the current location [31];
3. a value is randomly drawn from the conditional Gaussian distribution [35]:
$$Y(\mathbf{s}_0) \sim \mathcal{N}\left(\hat{Y}(\mathbf{s}_0), \sigma_K^2(\mathbf{s}_0)\right); \tag{5.10}$$
4. the simulated value is added to the conditioning dataset [54];
5. the process continues until all locations are simulated [23].

Unlike kriging, SGS preserves the histogram of the variable [36], the spatial covariance structure [31], realistic local variability [75], and spatial uncertainty patterns [54].

Multiple conditional realizations can be generated [35]:

$$\{Z^{(1)}(\mathbf{s}), \ldots, Z^{(m)}(\mathbf{s})\}, \tag{5.11}$$

allowing uncertainty propagation into downstream environmental or decision models [54].

Pointwise uncertainty can then be estimated empirically from the ensemble of realizations [23]:

$$\mathrm{Var}\left[Z(\mathbf{s})\right] \approx \frac{1}{m-1} \sum_{k=1}^{m} \left(Z^{(k)}(\mathbf{s}) - \bar{Z}(\mathbf{s})\right)^2. \tag{5.12}$$

Sequential Gaussian simulation is widely used in hydrogeology [54], reservoir characterization [36], mining [75], and climate applications [51] where extreme values and spatial heterogeneity are critical.

## 5.4 Sequential indicator simulation and non-Gaussian methods

Many environmental variables are non-Gaussian [105], multimodal [54], or strongly bounded [23].

Indicator approaches avoid strict Gaussian assumptions [54] by transforming the variable into threshold-based indicators [105]:

$$I(\mathbf{s}; z_c) = \begin{cases} 1, & Z(\mathbf{s}) \leq z_c, \\ 0, & Z(\mathbf{s}) > z_c. \end{cases} \tag{5.13}$$

Indicator kriging estimates conditional exceedance probabilities [55]:

$$P\left(Z(\mathbf{s}) \leq z_c \mid \mathcal{D}\right), \tag{5.14}$$

where $\mathcal{D}$ denotes the observed conditioning dataset [54].

Sequential Indicator Simulation (SIS) extends this framework to generate conditional realizations [54] without relying on Gaussianity [105].

At each location, values are sampled from locally estimated conditional cumulative distributions [54] reconstructed from multiple indicator thresholds [105].

These approaches are particularly useful when:

- **Distributions are highly skewed:** variables such as pollutant concentrations [55], mineral grades [105], rainfall [51], and disease incidence [54] often contain many low values and a few very large observations that are poorly represented by Gaussian assumptions.

- **Sharp boundaries exist:** categorical transitions [105], geological contacts [36], land-cover edges [54], and contamination fronts [55] may change abruptly across space, making smooth covariance-based interpolation inappropriate.

- **Threshold exceedances dominate the problem:** many decisions depend less on the exact predicted value than on whether a regulatory, ecological, or operational threshold is exceeded at a location [54].

- **Extreme events are physically important:** floods [51], droughts [8], heat waves [31], pollution peaks [55], and mineralized zones [36] may occupy small areas but have disproportionate consequences for risk assessment and decision-making.

Other non-Gaussian simulation methods include:

- **Plurigaussian simulation:** uses several latent Gaussian fields [105] to generate categorical or facies-like spatial patterns while preserving complex transitions between classes [36].

- **Multiple-point geostatistics:** learns spatial configurations from training images or conceptual patterns [54], allowing curvilinear, connected, or object-like structures to be reproduced more realistically [23].

- **Truncated Gaussian models:** transform an underlying Gaussian field into discrete categories by applying thresholds [105], which is useful for modeling ordered classes or domains separated by cutoffs [54].

- **Copula-based spatial simulation:** separates marginal distributions from dependence structure [8], making it possible to model non-Gaussian variables while retaining flexible spatial association [23].

Multiple-point geostatistics is especially important because it captures complex spatial patterns [54] that cannot be represented through two-point covariance functions alone [23].

## 5.5 Cokriging and multivariate spatial dependence

Many environmental variables are physically related across space [128]. Geostatistics can incorporate these relationships through multivariate covariance modeling [54].

Let

$$Z_1(\mathbf{s}), \ldots, Z_p(\mathbf{s}) \tag{5.15}$$

represent multiple spatial variables in a multivariate regionalized variable framework [128].

Cokriging exploits both direct covariance and cross-covariance structures [128]:

$$C_{ij}(\mathbf{h}) = \mathrm{Cov}\left[Z_i(\mathbf{s}), Z_j(\mathbf{s}+\mathbf{h})\right]. \tag{5.16}$$

The linear model of coregionalization represents these covariance functions as combinations of elementary spatial structures [128] and provides a practical framework for multivariate geostatistical prediction [54].

This framework is especially useful when:

- **The target variable is sparsely sampled:** for example, soil carbon [130], groundwater contamination [55], crop yield [65], or mineral assays [36] may be measured at only a limited number of field sites because sampling is expensive or destructive.
- **Secondary variables are densely available:** remote-sensing indices [65], elevation models [63], climate grids [51], geophysical surveys [128], or soil covariates [130] may cover the full study area and can help guide prediction between sparse target observations.
- **Multiple sensors observe related physical processes:** satellite imagery [67], field instruments [130], LiDAR [65], radar [51], weather stations [8], and laboratory measurements [54] may each capture different but correlated aspects of the same environmental system.

## 5.6 Block kriging and support effects

Spatial observations possess support [57]. A point sample [31], a raster pixel [65], and a regional average [32] are not equivalent quantities.

If support differences are ignored, prediction uncertainty may become artificially optimistic [57] and spatial interpretation may become invalid [32].

Block kriging addresses this problem by estimating averages over finite spatial regions [105] rather than point values [31].

Let

$$B \subset D \tag{5.17}$$

denote a spatial block or areal support [57]. The block prediction becomes:

$$\hat{Z}(B) = \frac{1}{|B|} \int_B \hat{Z}(\mathbf{s})\, d\mathbf{s}. \tag{5.18}$$

Because averaging reduces local variability [57], block kriging typically produces lower uncertainty than point kriging [105].

This distinction is particularly important in environmental applications where management decisions operate over finite spatial units [32] rather than mathematical points [57].

## 5.7 Indicator kriging and uncertainty modeling

Classical kriging methods often rely implicitly on Gaussian assumptions [31].

However, many environmental variables are strongly skewed [55], heteroscedastic [54], or heavy-tailed [105].

Indicator kriging addresses this issue by transforming the variable into binary indicators [54]:

$$I(\mathbf{s}; z_c) = \begin{cases} 1, & Z(\mathbf{s}) \leq z_c, \\ 0, & Z(\mathbf{s}) > z_c, \end{cases} \tag{5.19}$$

where $z_c$ is a threshold value defining the event of interest [105].

The resulting predictions correspond to exceedance probabilities [55] rather than point estimates [54].

This probabilistic interpretation is often more useful for risk analysis [54] and environmental decision support [55] than a single interpolated surface.

## 5.8 Spatiotemporal geostatistics

Many environmental systems evolve continuously in both space and time [8].

Spatiotemporal geostatistics extends covariance modeling [31] to joint space-time dependence [51]:

$$C(\mathbf{h}, u) = \mathrm{Cov}\left[Z(\mathbf{s}, t), Z(\mathbf{s} + \mathbf{h}, t + u)\right], \tag{5.20}$$

where $\mathbf{h}$ denotes spatial lag [54] and $u$ temporal lag [8].

Separable covariance models assume a product of spatial and temporal covariances [8]:

$$C(\mathbf{h}, u) = C_s(\mathbf{h})C_t(u), \tag{5.21}$$

although nonseparable formulations are often more realistic for dynamic environmental processes [51].

Spatiotemporal kriging and simulation are increasingly important in environmental monitoring and prediction [8]:

- **Meteorology:** estimating temperature [66], precipitation [56], wind speed [51], humidity [8], or storm intensity across space and time from weather stations, radar, and satellite observations.

- **Oceanography:** mapping sea-surface temperature [8], salinity [31], chlorophyll concentration [54], currents [51], or oxygen levels from sparse ship measurements, buoys, floats, and remote sensing products.

- **Hydrology:** predicting streamflow [8], groundwater levels [54], soil moisture [130], snowpack [51], or flood extent where observations are irregular and strongly influenced by catchment structure.

- **Climate reanalysis:** combining model output with observations [8] to reconstruct coherent historical fields of temperature, pressure, radiation, and moisture at regular spatial and temporal resolutions [51].

- **Air quality modeling:** estimating pollutant concentrations such as particulate matter [55], ozone [8], nitrogen dioxide [65], or smoke exposure by integrating monitoring stations, emissions data, meteorology, and satellite retrievals.

## 5.9 Strengths and limitations of geostatistics

Geostatistics is particularly strong when:

- **Spatial autocorrelation is dominant:** when nearby observations are strongly related [54], covariance and variogram models [31] can exploit this dependence to produce coherent interpolations and spatially informed predictions.

- **Uncertainty quantification is essential:** kriging variance [31], conditional simulation [35], and exceedance probabilities [55] provide explicit measures of confidence that are useful for risk assessment, monitoring, and decision support.

- **Sampling density is limited:** in field campaigns where observations are expensive [130], sparse [54], or unevenly distributed [31], geostatistical models can use spatial dependence to borrow information from neighboring locations.

- **Interpretability of spatial structure matters:** variogram range [54], sill [73], nugget [31], anisotropy [23], and nested components [128] provide physically meaningful summaries of scale, continuity, measurement noise, and directional dependence.

Its mathematical strength derives from the explicit modeling of covariance structure [116] and stochastic dependence [31].

However, classical geostatistics also possesses important limitations [23].

Most methods rely strongly on second-order stationarity [31] or intrinsic stationarity assumptions [92].

Under strong heterogeneity [8], abrupt discontinuities [54], or highly nonlinear covariate interactions [67], variogram estimation may become unstable or physically ambiguous [23].

In addition, covariance matrix inversion introduces computational limitations for very large datasets [116] because standard kriging scales approximately as

$$\mathcal{O}(n^3). \tag{5.22}$$

This cubic scaling follows from dense covariance linear algebra in standard Gaussian-process and kriging formulations [104].

Consequently, geostatistics alone may become insufficient when the dominant signal arises primarily from high-dimensional nonlinear predictor interactions [67] rather than spatial covariance itself [31].

# Chapter 6

# Machine learning for spatial prediction

## 6.1 Machine learning as nonlinear operator approximation

Machine learning methods approach spatial prediction differently from classical geostatistics [77]. Rather than modeling covariance structures explicitly [31], most machine learning algorithms attempt to learn nonlinear mappings [61] between predictors and responses [124]:

$$y_i = f(\mathbf{x}_i) + \eta_i, \tag{6.1}$$

where:

- $\mathbf{x}_i$ denotes predictor variables [61];
- $f$ is an unknown nonlinear operator [52];
- $\eta_i$ represents residual error [77].

From a functional analytic perspective, machine learning can therefore be interpreted as nonlinear operator approximation [52] in high-dimensional feature spaces [124].

In spatial applications, predictor variables may include:

- **Environmental covariates:** soil type [130], vegetation class [67], land use [65], geology [54], hydrological setting [8], or distance to rivers and roads [77] can provide contextual information about the process being modeled.
- **Topographic attributes:** elevation [63], slope [67], aspect [77], curvature [130], terrain wetness index [65], and drainage area [8] often explain spatial variation in temperature, erosion, soil moisture, biomass, and runoff.

- **Remote sensing variables:** spectral bands [67], vegetation indices [65], land-surface temperature [66], radar backscatter [77], canopy height [52], and texture metrics [84] can supply wall-to-wall predictors over large regions.

- **Climatic indicators:** precipitation [56], air temperature [66], solar radiation [8], evapotranspiration [65], wind exposure [51], drought indices [54], and seasonal summaries [8] can capture broad environmental forcing.

- **Spatial coordinates:** longitude [67], latitude [96], projected coordinates [77], or derived spatial basis functions [31] may help machine learning models represent broad location-dependent trends.

Importantly, including coordinates as predictors is not equivalent to explicitly modeling spatial covariance structure [31]; coordinate features can improve prediction [67] while still causing spatial validation concerns [96].

A model may exploit location information while still failing to represent stochastic spatial dependence probabilistically [31] or generalize under spatially structured validation [106].

## 6.2 Kernel methods and Gaussian processes

Several machine learning methods possess strong mathematical connections with geostatistics through kernel machines [113] and Gaussian process models [104].

Kernel methods define similarity through positive-definite kernels [113]:

$$K(\mathbf{x}_i, \mathbf{x}_j). \tag{6.2}$$

These kernels induce reproducing kernel Hilbert spaces [113] and allow high-dimensional nonlinear approximation through implicit feature mappings [49].

Gaussian process regression is particularly close to kriging because it models functions probabilistically [104] through covariance kernels [116]:

$$f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}')). \tag{6.3}$$

Under suitable assumptions, Gaussian process regression and kriging become mathematically equivalent up to differences in terminology [116], prior interpretation [104], and computational implementation [113].

## 6.3 Tree-based and ensemble methods

Tree-based methods are widely used in spatial prediction because they handle nonlinear interactions [12], heterogeneous predictors [67], and missing data effectively [11].

Random forests aggregate multiple decision trees [11]:

$$\hat{f}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} T_b(\mathbf{x}), \tag{6.4}$$

where $T_b$ denotes the $b$-th tree in the ensemble [11].

Gradient boosting methods sequentially minimize residual prediction error [45] through additive weak learners, as in modern boosted-tree implementations [22].

These methods often outperform classical geostatistical approaches when:

- **Many predictors are available:** models can exploit large feature sets from remote sensing [67], terrain analysis [63], climate grids [8], soil databases [130], land-cover products [65], and field measurements [54] without requiring a small predefined set of covariates.
- **Relationships are strongly nonlinear:** responses such as crop yield [65], vegetation stress [86], erosion risk [77], or pollutant concentration [55] may change abruptly once temperature, moisture, slope, or land-use thresholds are crossed.
- **Interactions are complex:** the effect of one variable may depend on another [61], such as elevation modifying temperature effects [66], soil texture modifying moisture response [130], or land cover modifying the relationship between rainfall and runoff [8].
- **Spatial autocorrelation is secondary relative to environmental controls:** in some settings, covariates such as elevation [63], vegetation index [65], distance to water [77], or climate [8] explain more variation than spatial proximity alone [67].

## 6.4 Deep learning and spatial-temporal representation

Deep learning extends machine learning to hierarchical nonlinear representation learning [86] through multilayer function approximation [52].

Convolutional neural networks are particularly useful for raster and image-based spatial prediction [84] because they exploit local spatial patterns directly [86].

Recurrent architectures support sequential dependence [52], while transformer architectures support attention-based spatial-temporal modeling [125].

More recently, graph neural networks have generalized these ideas toward irregular spatial domains [78], while neural operators extend learning toward mappings between functions [87] and operator learning frameworks [88].

## 6.5 Validation under spatial dependence

A major challenge in spatial machine learning is validation bias [106], especially when spatial sorting creates overly optimistic performance estimates [96].

Most machine learning theory assumes that training and testing samples are approximately independent [61] and similarly distributed [124].

Spatial autocorrelation violates this assumption because nearby observations are statistically related according to autocorrelation measures [97] and spatial covariance theory [31].

Consequently, random cross-validation often produces overly optimistic performance estimates in spatial ecology [106] and spatial prediction benchmarking [123].

Spatially blocked validation reduces this bias by separating training and testing regions geographically [100] or by using spatial blocking strategies [123].

This distinction is fundamental because a model may appear highly accurate under random validation [96] while failing to generalize to truly unobserved spatial regions [106].

## 6.6 Strengths and limitations of machine learning

Machine learning methods are particularly strong when:

- **Predictor dimensionality is large:** algorithms such as Random Forests [11], gradient boosting [45], and neural networks [52] can screen many covariates simultaneously, including remote-sensing bands [67], terrain metrics [63], climate summaries [8], and derived spatial features [77].

- **Nonlinear interactions dominate:** machine learning can represent threshold effects [61] and variable interactions [12], such as vegetation response depending jointly on rainfall [8], temperature [66], soil texture [130], and land management [65].

- **Large datasets are available:** modern models can benefit from millions of pixels [86], sensor records [67], or simulation outputs [52], making them useful for regional mapping [65], Earth Observation archives [84], and operational monitoring systems [8].

- **Predictive accuracy is prioritized:** when the main goal is minimizing error on new observations [61], flexible data-driven models often achieve strong performance by adapting to complex empirical patterns [77].

Their flexibility allows the representation of highly complex relationships [61] that are difficult to capture through classical covariance models [77].

Most conventional machine learning algorithms do not explicitly model spatial covariance structure [31] in the probabilistic sense used in geostatistics [54].

As a result:

- **Uncertainty quantification may be weak:** many algorithms produce point predictions [52] but do not naturally provide calibrated prediction intervals [48], exceedance probabilities [54], or spatially coherent uncertainty maps [31].

- **Spatial dependence may remain hidden in residuals:** a model can fit covariate relationships well [67] while still leaving clustered errors [106] that indicate unresolved spatial structure [31] or missing environmental drivers [54].

- **Extrapolation may become unstable:** predictions can be unreliable in locations with novel covariate combinations [96], land-cover types [65], elevations [63], or climates [8] that were poorly represented in the training data.

- **Validation procedures may become biased:** random cross-validation can overestimate performance [106] when training and test samples are spatially close [123], because the test set is not truly independent in space [100].

In environmental sciences, these limitations are especially important because uncertainty propagation [31] and spatial interpretability [54] are often as important as predictive accuracy itself [48].

# Chapter 7

# Spatial uncertainty and inverse problems

## 7.1 Prediction versus uncertainty

Spatial prediction alone is insufficient for scientific inference [31] because environmental decisions also require uncertainty characterization [54].

A map containing only expected values may appear visually convincing while concealing large uncertainty arising from sparse sampling [23], measurement noise [37], or model instability [8].

Consequently, spatial analysis should be interpreted as the joint problem of:

- estimating the spatial field [31];
- quantifying uncertainty [54];
- and characterizing the stability of the inference process [76].

This distinction becomes critical in environmental applications where sampling is irregular [31] and observations are expensive or incomplete [8].

## 7.2 Spatial prediction as an inverse problem

Spatial estimation can be interpreted mathematically as an inverse problem [39], with Bayesian inverse formulations [76] and geophysical inverse-problem interpretations [119].

Let

$$A : \mathcal{H} \to \mathbb{R}^n \tag{7.1}$$

denote an observation operator [39] mapping an unknown spatial field $f$ [31] into observations:

$$y = Af + \varepsilon. \tag{7.2}$$

The inverse problem consists of recovering $f$ from $y$ [7] under noise and incomplete observation constraints [76].

This problem is generally ill-posed because:

- observations are finite [39];
- measurements contain noise [119];
- spatial support may vary [57];
- multiple fields may explain the same observations [76].

Regularization is therefore essential in ill-posed inverse problems [120] and stable numerical reconstruction [39].

A common variational formulation is [76]:

$$\min_f \left[ \|Af - y\|^2 + \lambda \mathcal{R}(f) \right], \tag{7.3}$$

where:

- $\mathcal{R}(f)$ controls smoothness [1] or complexity [52];
- $\lambda$ balances data fidelity and regularization [120].

This formulation connects geostatistics [31], variational methods [39], Bayesian inference [76], Gaussian processes [104], and machine learning [52] through a common mathematical framework.

## 7.3 Regularization and spatial smoothness

Different regularization choices impose different assumptions concerning the geometry of spatial fields [39].

Tikhonov regularization penalizes large norms [120] to stabilize ill-posed problems [39]:

$$\mathcal{R}(f) = \|f\|^2. \tag{7.4}$$

Sobolev regularization incorporates weak derivatives [1] within function-space norms [13]:

$$\mathcal{R}(f) = \|\nabla f\|^2. \tag{7.5}$$

Total variation regularization instead preserves discontinuities [109] and sharp spatial transitions [76]:

$$\mathcal{R}(f) = \int_D |\nabla f(\mathbf{s})| \, d\mathbf{s}. \tag{7.6}$$

These choices are not merely computational details. They encode assumptions concerning smoothness [1], continuity [13], and spatial structure [76].

## 7.4 Uncertainty propagation in geostatistics

One major strength of geostatistics is explicit uncertainty representation through stochastic spatial models [31] and probabilistic mapping tools [54].

Kriging variance quantifies local prediction uncertainty in regionalized variable theory [92] and modern spatial prediction [31]:

$$\sigma_K^2(\mathbf{s}) = \mathrm{Var}\left[Z(\mathbf{s}) - \hat{Z}(\mathbf{s})\right]. \tag{7.7}$$

Importantly, this uncertainty depends not only on local sampling density [130] but also on the covariance structure of the field [116].

Conditional simulation extends this framework by generating multiple equiprobable spatial realizations [36] consistent with observations and covariance structure [23].

This allows:

- **Uncertainty propagation:** multiple realizations [36] can be passed through downstream models, such as hydrological [8], ecological [106], or exposure models [55], so that uncertainty in the spatial field is reflected in final outputs.

- **Risk assessment:** simulated maps [54] make it possible to identify areas where adverse outcomes, such as contamination [55], flooding [51], low yield [65], or habitat loss [106], remain plausible even if the mean prediction appears acceptable.

- **Probability estimation:** exceedance probabilities [54] can be computed for thresholds of practical interest, such as regulatory limits [55], drought levels [8], mineral cutoff grades [36], or safety standards [23].

- **Spatial decision analysis:** decision-makers can compare alternative sampling [130], remediation [55], conservation [106], or monitoring strategies [8] under many plausible spatial scenarios rather than relying on a single predicted map [54].

## 7.5 Uncertainty in machine learning

Uncertainty representation remains more difficult in many machine learning frameworks [48], especially when spatial dependence must also be represented [31].

Deterministic algorithms often provide point predictions [52] without probabilistic interpretation [48].

Several approaches attempt to address this limitation:

- Bayesian neural networks for approximate posterior uncertainty [46];
- ensemble learning for predictive distribution surrogates [85];
- quantile regression for conditional quantiles [79];
- conformal prediction for distribution-free coverage [127] and modern uncertainty sets [4];
- Gaussian process models for kernel-based probabilistic prediction [104].

However, uncertainty estimates from machine learning models may still fail to represent spatial covariance structure explicitly [106] or produce spatially calibrated uncertainty surfaces [48].

This distinction is important because predictive uncertainty and spatial dependence are not identical concepts [31]; a model can be uncertain without representing covariance, and it can exploit spatial patterns without calibrated uncertainty [54].

## 7.6 Sampling density and spatial reliability

The reliability of spatial inference depends strongly on sampling configuration [31] and field-survey design [130].

Sparse sampling [130] or clustered sampling [37] may fail to capture the true covariance structure of the field [30].

Under such conditions:

- variogram estimation becomes unstable [30];
- interpolation artifacts may arise [54];
- uncertainty estimates may become misleading [31].

This problem is especially severe in heterogeneous environments where multiple spatial scales coexist [128] and variogram structures become nested or scale dependent [23].

Consequently, visually smooth maps should never be interpreted as evidence of reliable spatial inference by themselves [54]; map smoothness can reflect model assumptions rather than true information content [31].

## 7.7 Transformation and variance stabilization

Environmental variables are frequently skewed [54], heteroscedastic [23], and heavy-tailed [36].

Because variograms depend on squared differences [31], extreme skewness may destabilize covariance estimation [54].

Transformations therefore play an important role in spatial modeling [54] and geostatistical simulation workflows [36].

Gaussian anamorphosis seeks a transformation [36] that stabilizes distributional assumptions for Gaussian geostatistical models [23]:

$$Y(\mathbf{s}) = \phi(Z(\mathbf{s})) \tag{7.8}$$

such that $Y(\mathbf{s})$ becomes approximately Gaussian [54].

Spatial modeling is then performed in transformed space [36] before back-transformation [75].

These procedures are not cosmetic statistical adjustments. They are often necessary for stable covariance modeling [54] and reliable uncertainty estimation [36].

# Chapter 8

# Support theory and multiscale spatial representation

## 8.1 Spatial support as part of the data definition

Spatial observations are not defined solely by their values [31] and locations [54]. They are also defined by their spatial support [57].

Support refers to the spatial region over which a measurement is effectively averaged [105], integrated [31], or observed [57].

A point sample [31], a raster pixel [57], a satellite footprint [50], and an administrative average [32] are fundamentally different spatial objects, even when they refer to the same environmental variable.

Let

$$B \subset D \tag{8.1}$$

denote a spatial support region [57]. The observed quantity may then be represented as:

$$Z(B) = \frac{1}{|B|} \int_B Z(\mathbf{s})\, d\mathbf{s}. \tag{8.2}$$

This formulation emphasizes that many observations are not exact point values, but spatial averages [105] over finite domains [57].

The importance of support is often underestimated in practical analysis. However, support directly affects:

- **Variance structure:** averaging over larger supports reduces local variability [105] and smooths short-range fluctuations [57].
- **Covariance behavior:** spatial dependence changes with support [31] because smoothing modifies correlation structure [32].
- **Interpretability:** values observed at different supports [50] do not necessarily represent the same physical quantity [57].

- **Prediction uncertainty:** uncertainty estimated at point scale [31] cannot automatically be transferred to block-scale prediction [105].

Ignoring support differences can generate:

- **Biased spatial comparisons:** point observations [31] may be incorrectly compared with areal averages [32] or coarse pixels [57].
- **Artificial smoothing:** support mismatch [50] can create the illusion of spatial continuity where none exists physically [54].
- **Overconfident uncertainty estimates:** uncertainty may be underestimated [57] when averaging effects are ignored [105].
- **Inconsistent multiscale predictions:** models calibrated at one support [32] may fail when transferred to another spatial resolution [17].

Consequently, support should be interpreted as a fundamental component of the measurement process [31] rather than a secondary technical detail [32].

## 8.2 The change-of-support problem

A fundamental consideration in spatial data analysis is the concept of *support*—the spatial, temporal, or spatiotemporal unit over which a variable is measured [50] or aggregated [57]. This concept becomes particularly important as data are collected from multiple sources, including drone imagery [114], proximal sensors [17], soil sampling [31], and satellite observations [33], at vastly different spatial resolutions. When information from these sources is combined [19] or compared [32], the differences in support can create misleading conclusions if not appropriately accounted for [17].

This issue, known as the **change of support problem (COSP)**, arises when spatial variables [57] or predictions [32] are represented at one resolution or scale, but are modeled or interpreted at another. For instance, high-resolution weed maps generated from UAV imagery [114] may be aggregated to the scale of agricultural management zones, while soil nutrient data may only be available at the field level [130] or as coarse interpolated maps from sparse sampling points [31]. When such datasets are fused without addressing the inherent difference in support [19], the resulting spatial estimates—whether of weed density, soil fertility, or treatment efficiency—can suffer from bias [115] and unquantified uncertainty [57].

In spatial analysis, techniques such as kriging [31], block kriging [105], spatial aggregation [57], or hierarchical Bayesian modeling [50] can be used to partially address the change of support problem [19]. Still, these methods come with their own assumptions [17] and limitations [32]. The choice of technique [19], resolution [57], and representation [8] directly influences the derived spatial patterns, model outputs, and ultimately the agronomic decisions made. Without careful handling, COSP can result in unjustified confidence in high-resolution maps [2] that are not statistically or biologically defensible [17].

This issue is pervasive in environmental sciences because most modern datasets are intrinsically multiscale [128] and often require nested or multiresolution geostatistical models [23].

Examples include:

- **Relating point soil samples to satellite pixels:** field observations represent local measurements [130], while remote sensing products represent spatial averages over larger footprints [33].

- **Combining station observations with climate grids:** weather stations provide point-scale information [8], whereas climate products often represent coarse regional averages [51].

- **Aggregating local measurements into management zones:** operational decisions are frequently made over finite spatial regions [57] rather than individual points [31].

- **Integrating heterogeneous remote sensing products:** different sensors may observe the same variable [59] at incompatible spatial resolutions [21] and supports [33].

Mathematically, changing support modifies both variance [57] and covariance structure [31].

Averaging over larger supports typically reduces variance [105]:

$$\mathrm{Var}[Z(B)] < \mathrm{Var}[Z(\mathbf{s})]. \tag{8.3}$$

At the same time, covariance structures generally become smoother and longer-ranged [54] because local fluctuations are partially averaged out [32].

These effects are not merely statistical artifacts. They reflect the fact that different supports observe different manifestations of the same latent spatial field [8].

Consequently:

- covariance models estimated at one support [31] cannot automatically be transferred to another support [57];

- variograms derived from coarse-resolution products [54] may fail to represent microscale variability [105];

- uncertainty estimates may become misleading [32] if support effects are ignored [17].

The change-of-support problem is therefore one of the central reasons why spatial prediction cannot be reduced to ordinary interpolation [31] without support-aware inference [57].

## 8.3 Block support and operational prediction

Many environmental decisions operate over finite regions [57] rather than mathematical points [31], especially when maps are used for applied management [54].

Agricultural management zones [114], ecological habitats [106], watersheds [8], administrative regions [32], and climate grid cells [51] are all examples of block support [57].

In such contexts, point prediction [31] may possess limited operational meaning because decisions are implemented over spatial aggregates [105].

Block kriging addresses this issue by estimating spatial averages [105] using geostatistical prediction principles [54]:

$$\hat{Z}(B) = \frac{1}{|B|} \int_B \hat{Z}(\mathbf{s})\, d\mathbf{s}. \tag{8.4}$$

This distinction is important because block prediction and point prediction possess different statistical properties [31] because averaging changes the target variable [57].

Block prediction generally exhibits:

- **Reduced variance:** averaging suppresses local stochastic variability [105].

- **Greater stability:** predictions become less sensitive to isolated observations [31] or local anomalies [54].

- **Improved operational relevance:** predictions align more closely with the spatial scale of management [114] or policy decisions [57].

- **Different uncertainty behavior:** block uncertainty reflects aggregated variability [32] rather than pointwise fluctuation [31].

This distinction is operationally important because visually precise point maps may suggest unrealistic levels of local certainty [54] when the relevant support is areal [57].

In many environmental applications, the relevant inference problem is not:

$$Z(\mathbf{s}), \tag{8.5}$$

but rather:

$$Z(B). \tag{8.6}$$

Support-aware prediction is therefore essential for scientifically defensible decision-making [57] and probabilistic spatial analysis [8].

## 8.4 Scale dependence and spatial organization

Environmental systems are intrinsically multiscale [128] and often require geostatistical models with multiple spatial structures [23].

Observed spatial variability often reflects the superposition of:

- **Local stochastic variability:** short-range heterogeneity [54] associated with microscale processes [105] or measurement noise [31].
- **Intermediate-scale environmental gradients:** variability associated with terrain [63], hydrology [8], vegetation [106], or land use [65].
- **Large-scale deterministic forcing:** broad spatial structure generated by climate [51], geology [54], or regional circulation patterns [8].

A single covariance structure may therefore fail to represent all relevant spatial scales adequately [128] or preserve local spatial continuity [54].

Nested variogram models address this issue by representing covariance through multiple spatial structures [128], a principle also exploited in factorial kriging [53]:

$$\gamma(\mathbf{h}) = \sum_{k=1}^{K} \gamma_k(\mathbf{h}), \tag{8.7}$$

where each component corresponds to a distinct spatial scale [128].

This decomposition is important because different physical mechanisms often dominate at different scales [128].

For example:

- **Microscale variability** may reflect local soil heterogeneity [130], plant-scale ecological variation [106], measurement noise [31], or unresolved turbulence [51] that changes over distances smaller than the sampling interval.

- **Intermediate scales** may reflect hillslope processes [8], watershed structure [54], land-management units [114], vegetation patches [106], or ecological organization that creates coherent patterns over tens to thousands of meters.

- **Large scales** may reflect regional climatic forcing [51], geological structure [54], latitude [31], elevation gradients [63], or broad land-use transitions [65] that impose smooth spatial trends across the entire study domain.

Scale dependence strongly influences:

- **Predictive performance:** a model may predict well at coarse resolution [57] but fail to capture local extremes [54], or it may fit fine-scale noise while missing regional gradients [128].

- **Uncertainty quantification:** prediction uncertainty depends on which scales are sampled [31], which scales are modeled [23], and which scales remain unresolved by the observation network [130].

- **Physical interpretability:** separating short-, medium-, and long-range components [53] helps connect statistical patterns to mechanisms such as terrain control [63], hydrological connectivity [8], or climate forcing [51].

- **Transferability across regions:** relationships learned at one scale [128] or in one landscape may not generalize elsewhere if the dominant spatial processes differ between regions [106].

A model that performs well at one scale may fail at another because the dominant physical processes change with scale [128].

Understanding multiscale organization is therefore central to both scientific interpretation [54] and predictive modeling [23].

## 8.5 Wavelets and multiresolution analysis

Functional analysis provides additional tools for multiscale spatial representation through wavelets [34] and multiresolution decompositions [90].

Wavelet expansions represent spatial fields as:

$$f(\mathbf{s}) = \sum_{j,k} c_{jk}\psi_{jk}(\mathbf{s}), \tag{8.8}$$

where:

- $\psi_{jk}$ are localized basis functions representing spatial patterns [34];

- $j$ controls scale or resolution [90];
- $k$ controls spatial location [98].

Unlike Fourier methods, wavelets preserve both:

- **Spatial localization:** local features remain spatially identifiable [34];
- **Scale information:** different resolutions can be analyzed simultaneously [90].

This property is especially useful for:

- **Nonstationary spatial fields:** local statistical properties may vary across the domain [98].
- **Abrupt discontinuities:** sharp transitions are difficult to represent with globally smooth basis functions [34].
- **Multiscale environmental systems:** variability may emerge from processes acting at very different ranges [128].
- **Spatial-temporal variability:** localized transient structures can be represented efficiently [90].

Wavelet methods additionally provide:

- data compression [34];
- noise filtering [98];
- multiscale decomposition [90];
- sparse functional representation [34].

These methods establish important connections between spatial statistics [98], harmonic analysis [34], signal processing [90], and modern machine learning representations [52].

## 8.6 Functional representations and spatial complexity

Functional representations become increasingly important as spatial datasets grow in dimensionality [102] and complexity [69].

Environmental datasets frequently involve:

- **Hyperspectral signatures:** observations consist of high-dimensional spectral curves [33] rather than scalar values [102].

- **Temporal trajectories:** variables evolve continuously through time as spatial-temporal functions [69].

- **Multiscale raster stacks:** information is distributed across many spatial resolutions [21] and covariate layers [67].

- **Spatial-temporal fields:** environmental systems evolve jointly across space and time [8].

- **Coupled physical processes:** interacting systems generate complex multivariate dependence structures [128].

In these situations, observations are often more naturally interpreted as functions [69] or operators [82] rather than finite-dimensional vectors.

Functional data analysis extends classical statistics by representing observations in infinite-dimensional spaces through principal component representations [103], functional data analysis [102], and stochastic process theory [69].

This framework supports:

- **Functional regression:** relationships are modeled between curves [102], surfaces [69], or fields [31] rather than scalar variables.

- **Operator learning:** mappings between functional spaces can be approximated directly [82].

- **Spatial-temporal decomposition:** complex variability can be separated across scales [128] and dimensions [69].

- **Multiscale regularization:** smoothness [102] and complexity constraints [52] can be imposed across multiple resolutions simultaneously [90].

These ideas are increasingly relevant because environmental systems are heterogeneous [31], multiscale [128], partially observed [8], and dynamically coupled [82].

From this perspective, spatial representation becomes not merely a problem of storing coordinates and values, but a broader problem of describing complex spatial structure within infinite-dimensional functional spaces [110] and stochastic function spaces [69].

# Chapter 9

# Data fusion and multivariate spatial analysis

## 9.1 Why spatial data fusion is necessary

Environmental systems are rarely observed through a single variable [59] or sensor [33].

Different observation systems provide complementary information at different resolutions [21], supports [57], accuracies [107], temporal frequencies [8], and spatial extents [33].

Examples include:

- **Satellite imagery:** provides large spatial coverage [33] and repeated observations [21], but often with indirect measurements [59], atmospheric contamination [3], or coarse support [57].

- **Field measurements:** offer direct and usually more accurate local observations [130], but are sparse [31], expensive [54], and irregularly distributed [8].

- **Climate reanalysis products:** combine physical models and observations [8] to generate spatially complete fields, although they may contain systematic biases inherited from model assumptions [48].

- **Geophysical surveys:** provide indirect information about subsurface [119] or environmental structure [39] through electromagnetic, seismic, or radiometric responses.

- **Topographic variables:** encode terrain-driven controls such as slope [63], curvature [130], drainage [8], and exposure [77], which strongly influence many environmental processes.

- **Numerical simulation outputs:** represent physically constrained system behavior [82] but may differ from observations because of parameter uncertainty [76] or model simplification [119].

No single dataset fully characterizes the spatial process [59] or resolves all reliability limitations [107].

Data fusion therefore seeks to combine heterogeneous information sources into a coherent spatial representation [59], using multisensor fusion principles [18] and modern data-integration strategies [3].

This problem is mathematically challenging because the datasets may differ in:

- **Support:** measurements may represent points [31], pixels [33], blocks [57], or integrated volumes [119], which are not directly comparable.
- **Scale:** spatial resolution [21] and effective averaging scale [105] may differ substantially across sensors.
- **Uncertainty:** different observation systems possess distinct noise structures [76], calibration errors [107], and reliability levels [59].
- **Sampling density:** some variables may be densely available [33] while others are sparse [31] or irregular [130].
- **Physical interpretation:** sensors may observe different manifestations [59] of the same underlying process rather than the process itself [8].

Consequently, data fusion should not be interpreted merely as stacking datasets together. It is fundamentally the problem of integrating multiple incomplete observations [31] of a latent spatial field [8].

An ideal method for addressing COSP in data fusion [19] should: (i) explicitly account for differences in spatial support [57]; (ii) allow both upscaling and downscaling of data [32]; (iii) incorporate measurement uncertainties in predictions [8]; (iv) integrate diverse covariates [59]; (v) preserve consistency across scales, including mass balance in upscaling and pycnophylactic properties in downscaling [50]; and (vi) be practical for implementation in a GIS environment for various support transformations [17].

Addressing COSP effectively is crucial for reliable data fusion [19], particularly when predictions are needed at scales different from those of the original measurements [57]. However, finding a universal statistical method that fully meets the complex requirements of COSP remains a challenging task [17]. Castrignanò and Buttafuoco provide a comprehensive review of current methods used to address this problem [19], each with its own strengths and limitations [32]. For this discussion, I arise three methods. Table 9.1 compares examples of statistical approaches to data fusion and the COSP. The following subsections briefly describe these statistical approaches that have been developed to tackle these issues.

**Table 9.1.** Comparison of Statistical Approaches to Data Fusion and Change of Support Problem

| Method | Support Handling | Scale Direction | Uncertainty Treatment | GIS Integration Capability |
|---|---|---|---|---|
| Geographically Weighted Regression (GWR) | Implicitly through spatial weighting kernels | Primarily downscaling | Limited; error estimates local but not probabilistic | Good, point-to-point or point-to-area via local models |
| Multiscale Spatial Tree Models | Hierarchical partitions support multi-resolution inference | Both upscaling and downscaling | Moderate; requires post hoc uncertainty estimation | Moderate; integration requires custom implementation |
| Bayesian Hierarchical Models | Explicit through latent spatial process modeling | Both upscaling and downscaling | Strong; uncertainty is fully propagated | Moderate to high; implementation via external tools like R-INLA or Stan |
| Multivariate Geostatistics (e.g., cokriging) | Explicit using variogram models and kriging equations | Both; area-to-point and block predictions supported | Moderate to strong; kriging variance available | Good; well-supported in GIS geostatistics toolkits |

## 9.2 Multivariate spatial dependence

Let

$$Z_1(\mathbf{s}), \ldots, Z_p(\mathbf{s}) \tag{9.1}$$

denote multiple spatial variables in multivariate geostatistics [128] and applied cokriging [54].

Multivariate spatial analysis seeks to model both:

- **Direct spatial dependence:** the autocorrelation structure of each individual variable across space [31].
- **Cross-dependence between variables:** statistical relationships linking different variables [128] through shared spatial organization or common physical controls [41].

Cross-covariance functions are defined as [128]:

$$C_{ij}(\mathbf{h}) = \mathrm{Cov}\left[Z_i(\mathbf{s}), Z_j(\mathbf{s}+\mathbf{h})\right]. \tag{9.2}$$

These structures allow information from secondary variables [54] to improve prediction of sparsely sampled targets [31].

This principle is especially valuable when:

- **Direct sampling is expensive:** some environmental variables require costly laboratory analysis [130] or difficult field acquisition [54].
- **Auxiliary variables are densely available:** remote sensing products [33] or terrain attributes [63] may provide nearly continuous spatial coverage.
- **Variables share common physical controls:** correlated processes [128] may reveal latent environmental mechanisms that are not directly observable through a single variable alone [41].

For example, soil moisture, vegetation indices, topography, and temperature may all exhibit coupled spatial structure [54] because they respond jointly to hydrological [8] and climatic forcing [51].

Multivariate modeling therefore allows the analyst to exploit both within-variable continuity [128] and between-variable relationships [41].

## 9.3 The linear model of coregionalization

A classical framework for multivariate geostatistics is the linear model of coregionalization (LMC) [128].

The LMC represents covariance structures [31] as combinations of shared elementary spatial components [128]:

$$C_{ij}(\mathbf{h}) = \sum_{k=1}^{K} b_{ij}^{(k)} g_k(\mathbf{h}), \tag{9.3}$$

where:

- $g_k(\mathbf{h})$ are elementary covariance structures [128] associated with specific spatial scales or spatial processes [53];
- $b_{ij}^{(k)}$ are coregionalization coefficients [128] controlling how strongly each variable participates in each spatial structure [41].

This decomposition permits:

- **Multiscale covariance modeling:** different spatial ranges [128] can be represented simultaneously within the same framework [53].
- **Cross-variable interpolation:** information from correlated variables [41] can improve prediction where direct observations are sparse [54].

- **Spatial process decomposition:** distinct physical mechanisms [8] may be separated according to their characteristic spatial scales [128].

The LMC is particularly important because environmental systems rarely operate through a single spatial mechanism. Large-scale climate forcing [51], intermediate-scale terrain effects [63], and local stochastic heterogeneity [54] may coexist simultaneously.

The framework therefore provides both predictive capability and physical interpretability [128].

## 9.4 Factorial kriging and scale decomposition

Factorial kriging extends the LMC by decomposing total spatial variability into approximately independent spatial components [53] within the multivariate geostatistical framework [128].

A spatial field may then be represented as:

$$Z(\mathbf{s}) = \sum_{k=1}^{K} Y_k(\mathbf{s}), \tag{9.4}$$

where each component $Y_k(\mathbf{s})$ corresponds to a distinct spatial scale [128].

This decomposition supports:

- **Separation of short-range and long-range variability:** local noise [31] can be distinguished from broad regional structure [128].
- **Identification of dominant spatial scales:** different physical mechanisms [53] may dominate at different ranges [54].
- **Interpretation of physical driving mechanisms:** scale decomposition [128] often reveals meaningful environmental organization [41].
- **Multiscale environmental analysis:** different scales can be analyzed independently depending on the scientific question [53].

For example:

- microscale variability may reflect local heterogeneity [130] or measurement noise [31];
- intermediate scales may reflect geomorphology [54] or land-use patterns [65];
- large scales may reflect climatic [51] or geological forcing [8].

Factorial kriging is therefore useful not only for prediction, but also for scientific interpretation [53] and process understanding [128].

## 9.5 Geographically Weighted Regression

Geographically Weighted Regression (GWR) is a local regression technique [16] that allows model coefficients to vary across geographic space [43]. This flexibility enables the model to capture non-stationary relationships [95] between a dependent variable and its predictors, which is useful in precision agriculture [58] where local field conditions often lead to spatially varying agronomic processes [27].

$$y(\mathbf{s}_i) = \beta_0(\mathbf{s}_i) + \sum_{k=1}^{p} \beta_k(\mathbf{s}_i) x_k(\mathbf{s}_i) + \varepsilon(\mathbf{s}_i) \tag{9.5}$$

In equation (9.5), $y(\mathbf{s}_i)$ represents the target variable at location $\mathbf{s}_i$ [16], $x_k(\mathbf{s}_i)$ are the predictor variables [43], $\beta_k(\mathbf{s}_i)$ are spatially varying regression coefficients [95], and $\varepsilon(\mathbf{s}_i)$ is a random error term [40].

Haghighattalab et al. used GWRs to predict grain yield on a plot level [58] by examining relationships between unmanned aerial systems imagery and grain yield. They found high correlations between measured phenotypic traits derived from imagery and grain yield in drought and irrigated environments [58]. Residuals from GWR models were lower and less spatially dependent than methods using principal component regression, suggesting the value of spatially corrected models [58]. Evans et al. investigated the use of GWR for analysis of data from large on-farm experiments [40].

## 9.6 Multiscale Spatial Tree Models

Multiscale Spatial Tree Models decompose the spatial domain hierarchically into nested regions [70] and fit predictive models at each level of the hierarchy [81]. These models support scale transitions by aggregating or disaggregating local predictions across tree levels [72]. Their structure makes them suitable for applications where both global trends [42] and local variations [131] must be represented simultaneously.

$$y(\mathbf{s}) = \sum_{m \in \mathcal{T}} \phi_m(\mathbf{s}) \theta_m + \varepsilon(\mathbf{s}) \tag{9.6}$$

Here, $y(\mathbf{s})$ is the observed value at location $\mathbf{s}$ [70], $\phi_m(\mathbf{s})$ are spatial basis functions defined over tree node $m \in \mathcal{T}$ [72], $\theta_m$ are scale-specific coefficients [81], and $\varepsilon(\mathbf{s})$ is the model error [42].

Zhu et al. applied the multi-scale method by combining soil coring, penetrometer, and topographic data [131] to map depth-to-till at 10-m resolution and field capacity at 20-m resolution. They evaluated different combinations of datasets, finding that when three data layers were present, the multi-scale model performed slightly better than regression kriging and cokriging [131]. When two data layers were present, the multi-scale model produced results similar to regression kriging, but both outperformed cokriging, probably because soil core measurements were too sparse for cokriging to perform well [131]. In situations where more than two data sources are available for mapping, the multi-scale model showed better results [131].

## 9.7 Bayesian data fusion and hierarchical modeling

Bayesian hierarchical models provide a flexible probabilistic framework for spatial data fusion [8] and spatial statistical inference [31].

A hierarchical formulation separates:

- **The latent spatial process:** the unobserved environmental field that one ultimately seeks to infer [8].
- **The observation process:** the mechanism through which sensors or measurements observe the latent field [50].
- **Uncertainty propagation:** probabilistic treatment of noise [76], measurement error [31], and incomplete sampling [8].

A generic formulation is:

$$y_i = A_i f + \varepsilon_i, \tag{9.7}$$

where:

- $f$ is the latent spatial field [8];
- $A_i$ is an observation operator describing how sensor $i$ measures the field [39];
- $\varepsilon_i$ denotes sensor-specific noise or measurement uncertainty [76].

This framework naturally accommodates:

- **Heterogeneous sensors:** different observation systems may possess distinct resolutions [21] and error structures [59].
- **Multiscale support:** measurements collected at different supports [57] can be represented consistently through observation operators [50].
- **Missing observations:** incomplete sampling [31] can be incorporated probabilistically rather than treated as a purely technical inconvenience [8].

- **Uncertainty propagation:** posterior distributions provide uncertainty estimates [8] directly linked to both observations and model assumptions [50].

Bayesian fusion methods are increasingly important in climate science, remote sensing, and environmental forecasting because they provide a coherent framework for integrating physical knowledge and uncertain observations simultaneously [8] and for spatiotemporal fusion problems [99]. Bayesian Hierarchical Models (BHMs) provide a coherent probabilistic framework for modeling spatial data collected at multiple resolutions [80]. These models explicitly account for uncertainties [8] and allow for the integration of diverse data sources by separating the data level, the process level, and the parameter level [118].

$$y_i \sim \mathcal{N}(z(\mathbf{s}_i), \sigma_i^2) \tag{9.8}$$

$$z(\mathbf{s}) = \mu(\mathbf{s}) + w(\mathbf{s}) \tag{9.9}$$

$$\mu(\mathbf{s}) = \mathbf{x}(\mathbf{s})^\top \boldsymbol{\beta}, \quad w(\mathbf{s}) \sim \mathrm{GP}(0, C(\mathbf{s}, \mathbf{s}')) \tag{9.10}$$

In equations (9.8)–(9.10), $y_i$ are the observations [80], $z(\mathbf{s})$ is the latent spatial process [8], $\mu(\mathbf{s})$ is a mean function determined by covariates $\mathbf{x}(\mathbf{s})$ and coefficients $\boldsymbol{\beta}$ [31], $w(\mathbf{s})$ is a spatially correlated random effect [118], and $\sigma_i^2$ is the observational error variance [76].

Stojanović et al. evaluated the association between leaf area index and spectral reflectance at various wavelengths using spline-based kernel functions embedded into Bayesian Hierarchical Models [118]. They compared models with three different levels of hierarchy: a non-hierarchical baseline model, a model with a hierarchical bias parameter, and a model in which bias and kernel parameters are hierarchically structured [118]. The authors considered Bayesian Hierarchical Models to yield qualitative benefits regarding interpretability, model complexity, and robustness [118].

## 9.8 Machine learning for spatial fusion

Machine learning methods have substantially expanded the capabilities of spatial data fusion [21], especially for remote-sensing and hyperspectral data integration [33].

Deep learning architectures can integrate:

- **Raster imagery:** convolutional structures [86] can extract spatial patterns from satellite or aerial data [33].
- **Spatial-temporal sequences:** recurrent architectures [52] and transformer architectures [125] can model evolving dynamics across space and time.

- **Graph structures:** graph neural networks support irregular spatial relationships [78] and network-based systems [59].
- **Physical simulations:** numerical model outputs [82] can be incorporated jointly with observations [76].
- **Heterogeneous sensor streams:** multimodal architectures can combine distinct forms of spatial information [18] simultaneously [3].

Attention mechanisms and transformer architectures additionally permit adaptive weighting of multiple information sources according to their contextual relevance [125].

These methods are powerful because they can learn highly nonlinear relationships [52] without requiring explicit covariance assumptions [86].

However, purely data-driven fusion methods may still struggle with uncertainty calibration [48] and spatial generalization [106]:

- **Uncertainty quantification:** many deep models provide point predictions [52] without rigorous probabilistic interpretation [48].
- **Physical interpretability:** learned representations may be difficult to connect with known environmental mechanisms [8].
- **Extrapolation outside training conditions:** predictive performance may degrade substantially in unseen spatial [106] or climatic regimes [51].
- **Sparse observational regimes:** deep learning models may require more data [86] than are available in many environmental applications [31].

These limitations motivate hybrid frameworks [20] combining geostatistical and machine-learning perspectives [77]:

- physical models [82];
- geostatistics [31];
- probabilistic inference [8];
- machine learning [52].

## 9.9 Data fusion as operator integration

From a functional analytic perspective, data fusion can be interpreted as the integration of multiple observation operators acting on a common latent spatial field through inverse-problem operators [39] and Bayesian inverse modeling [76].

Let

$$A_1, \ldots, A_m \tag{9.11}$$

represent distinct sensing operators in data fusion [59] and inverse-problem theory [119].

The fused inverse problem becomes:

$$y_i = A_i f + \varepsilon_i, \quad i = 1, \ldots, m. \tag{9.12}$$

This formulation is conceptually important because different datasets do not observe the same object in the same way [57] and may correspond to different statistical observation processes [31].

Each operator may:

- sample different spatial scales [128];
- introduce different smoothing effects [105];
- possess different uncertainty structures [76];
- observe different physical manifestations of the latent field [8].

This viewpoint unifies:

- **Multivariate geostatistics:** covariance-based integration of multiple spatial variables [128];
- **Inverse problems:** reconstruction of latent fields from indirect observations [39];
- **Bayesian hierarchical models:** probabilistic integration of heterogeneous information sources [8];
- **Machine learning fusion architectures:** nonlinear approximation of complex multimodal relationships [52].

Consequently, spatial data fusion is not merely a practical procedure for combining datasets. It is fundamentally an operator-theoretic problem defined over multiscale functional spaces [39] and stochastic function-space representations [69].

This interpretation becomes increasingly important as environmental datasets grow in dimensionality, heterogeneity, and spatial-temporal complexity [33] and multiresolution structure [21].

# Chapter 10

# Hybrid geostatistics and machine learning

## 10.1 Beyond the false dichotomy

Geostatistics and machine learning are often presented as competing approaches to spatial prediction. This interpretation is misleading because geostatistics addresses stochastic spatial dependence [31], machine learning addresses nonlinear prediction [77], and regression kriging combines both components [64].

Both approaches seek to infer unknown spatial functions from incomplete observations, but they emphasize different mathematical structures, including geostatistical covariance information [54] and machine-learning predictor information [61].

Geostatistics focuses primarily on:

- **Stochastic spatial dependence:** nearby observations are treated as statistically related through covariance [31] or variogram structures [54]. The spatial configuration of the data is therefore part of the model itself [92].

- **Covariance structure:** spatial continuity is modeled explicitly [116], allowing the analyst to characterize spatial scales [128], anisotropy [54], and local uncertainty [31].

- **Uncertainty quantification:** prediction is accompanied by probabilistic information such as kriging variance [31] or conditional realizations [54], which are essential for risk-sensitive applications [8].

- **Probabilistic interpolation:** predictions are derived from stochastic assumptions concerning the spatial field [92] rather than only from empirical predictive accuracy [61].

Machine learning emphasizes:

- **Nonlinear approximation:** flexible algorithms can capture highly nonlinear relationships [61] between predictors and responses without requiring explicit parametric assumptions [52].
- **High-dimensional predictors:** modern ML methods can integrate large numbers of environmental variables [77], raster layers [33], temporal features [125], and remote sensing products [67] simultaneously.
- **Representation learning:** deep architectures can automatically extract latent structures [86] and hierarchical features from complex spatial inputs [52].
- **Predictive scalability:** many ML methods remain computationally practical [11] for datasets that are too large for classical covariance-based methods [116].

From a functional analytic perspective, these approaches are not mutually exclusive. They correspond to complementary approximation strategies acting on spatially structured function spaces, including Gaussian-process kernels [104] and RKHS methods [113].

This observation is important because many environmental systems contain both deterministic nonlinear structure and latent stochastic spatial dependence. Ignoring either component may produce incomplete inference, as shown in regression-kriging workflows [63] and hybrid geostatistical learning studies [44].

This complementarity motivates regression kriging [63], spatial prediction with covariates [68], machine-learning residual kriging [44], and random-forest kriging applications [126].

## 10.2 Residual spatial structure and decomposition

A central idea in hybrid modeling is that deterministic predictors rarely explain all spatial variability in regression kriging [64] or spatial random-forest models [67].

Let

$$Z(\mathbf{s}) = f(\mathbf{x}(\mathbf{s})) + \varepsilon(\mathbf{s}), \tag{10.1}$$

where:

- $f(\mathbf{x}(\mathbf{s}))$ captures deterministic nonlinear relationships [61] between predictors and the target variable [67];

- $\varepsilon(\mathbf{s})$ represents residual spatial dependence [31] that remains unexplained after accounting for the predictors [64].

This decomposition is conceptually important because environmental systems are usually driven by multiple mechanisms simultaneously, including local spatial continuity [54] and multiscale variability [128].

Part of the variability may be explained by observable covariates such as:

- **Elevation:** controls temperature [8], pressure, drainage [54], erosion potential [77], and exposure, so it often explains broad gradients in environmental variables [63].

- **Land cover:** distinguishes forests, croplands, urban areas, wetlands, bare soil, and water bodies [33], each of which may influence reflectance [67], evapotranspiration, runoff, and local microclimate [8].

- **Soil properties:** texture [130], organic matter, pH, bulk density, drainage class, and mineral composition can affect water retention, nutrient cycling, contaminant mobility, and vegetation response [54].

- **Climate variables:** precipitation [51], temperature [8], solar radiation, humidity, wind, and evapotranspiration summarize broad environmental forcing that shapes spatial patterns over seasonal to long-term timescales [31].

- **Remote sensing products:** spectral bands [33], vegetation indices [67], land-surface temperature, radar backscatter, and canopy metrics provide spatially continuous information [65] that can reveal patterns not captured by sparse field samples [130].

Another component may arise from latent spatial processes that are not directly observed, including:

- **Unresolved environmental heterogeneity:** fine-scale variation in soils [130], vegetation [106], hydrology [8], or management history may remain invisible because it occurs below the resolution of available covariates [57].

- **Missing predictors:** important drivers such as groundwater depth [54], pollutant sources, local disturbance, or historical land use may not be measured, leaving spatial structure in the residuals [31].

- **Local transport mechanisms:** wind [51], water flow [8], sediment movement, animal dispersal [106], or human infrastructure can redistribute material across space in ways not captured by static covariates.

- **Measurement aggregation effects:** point samples [31], pixels [33], plots, and administrative units [57] represent different spatial supports, so averaging or resampling can introduce residual patterns [32].

- **Multiscale stochastic variability:** random variation may occur simultaneously at local, landscape, and regional scales [128], requiring residual models that can represent more than one spatial scale [23].

Machine learning models may approximate the deterministic component effectively while still leaving residual spatial autocorrelation [106] or spatially structured prediction bias [96].

Geostatistics can then model the remaining stochastic structure through covariance analysis [31] and variogram modeling [54].

This decomposition is mathematically important because it separates:

- **Nonlinear covariate effects:** deterministic variation explained by observable environmental controls [61];

- **Latent spatial dependence:** structured residual variability associated with unresolved spatial organization [31];

- **Uncertainty propagation:** probabilistic characterization of the unexplained component of the field [54].

This interpretation also clarifies why adding coordinates directly as predictors is not equivalent to modeling covariance explicitly. Coordinates may help prediction numerically while still failing to provide a coherent stochastic representation of spatial dependence [31] or spatial transferability [106].

## 10.3 Regression kriging and residual kriging

One of the most widely used hybrid methods is regression kriging [64], also described as spatial prediction with regression and residual kriging [68].

The prediction becomes:

$$\hat{Z}(\mathbf{s}) = \hat{f}_{\mathrm{ML}}(\mathbf{x}(\mathbf{s})) + \hat{\varepsilon}_{\mathrm{OK}}(\mathbf{s}), \tag{10.2}$$

where:

- $\hat{f}_{\mathrm{ML}}$ is a machine learning predictor responsible for capturing deterministic structure [67];

- $\hat{\varepsilon}_{\mathrm{OK}}$ is the kriged residual field, responsible for modeling remaining spatial autocorrelation [64].

This formulation combines:

- **Nonlinear predictive flexibility:** machine learning captures complex interactions [61] that may be difficult to express through classical parametric models [52];
- **Explicit spatial covariance modeling:** kriging preserves information about spatial continuity [31] and local dependence [54];
- **Probabilistic uncertainty representation:** residual kriging provides interpretable uncertainty estimates [64] linked to spatial configuration and covariance structure [31].

Residual kriging is especially effective when:

- **Environmental covariates explain large-scale variation:** broad deterministic trends can be learned through regression [68] or machine learning components [67];
- **Residual spatial autocorrelation persists locally:** remaining dependence can then be modeled separately through geostatistical methods [31];
- **Uncertainty quantification remains operationally important:** risk-sensitive applications require more than point prediction alone [54].

In practice, regression kriging often improves both predictive accuracy and spatial realism because the deterministic and stochastic components are modeled separately rather than forced into a single framework [44].

## 10.4 Gaussian processes and kernel unification

Gaussian process regression provides one of the clearest theoretical bridges between geostatistics [116] and machine learning [104].

A Gaussian process defines a probability distribution over functions [104]:

$$f(\mathbf{x}) \sim \mathcal{GP}\left(m(\mathbf{x}), K(\mathbf{x}, \mathbf{x}')\right). \tag{10.3}$$

The covariance kernel

$$K(\mathbf{x}, \mathbf{x}') \tag{10.4}$$

plays the same mathematical role as covariance functions in geostatistics [31] and Gaussian process regression [104].

Under appropriate assumptions, Gaussian process regression and kriging become equivalent formulations differing mainly in:

- **Terminology:** geostatistics typically emphasizes variograms [54] and stochastic fields [31], while machine learning emphasizes kernels [113] and probabilistic function learning [104];

- **Prior interpretation:** Gaussian processes are usually framed within Bayesian inference [104], whereas kriging historically emerged from regionalized variable theory [92];

- **Computational implementation:** the underlying mathematics is closely related [116], but optimization and software ecosystems often differ substantially [113].

This equivalence is theoretically important because it demonstrates that covariance-based geostatistics and kernel-based machine learning share Hilbert-space foundations in positive-definite kernels [6], smoothing splines [129], and kernel machine learning [113].

Kernel methods, kriging, and probabilistic machine learning therefore emerge as different expressions of a common approximation framework [104].

## 10.5 Spatial deep learning and operator learning

Recent developments extend hybridization beyond classical residual modeling toward Fourier neural operators [87] and modern operator learning [82].

Neural operators learn mappings between functional spaces [87] rather than finite-dimensional vectors [82], while related operator-learning architectures approximate nonlinear operators directly [88].

This distinction is important because many environmental systems are naturally governed by partial differential equations [101]:

$$\mathcal{L}u = f. \tag{10.5}$$

In these settings, the objective is not merely interpolation between points, but approximation of spatial-temporal dynamics constrained by physical laws [101] and learned solution operators [87].

Examples include:

- **Atmospheric transport:** advection, diffusion, and turbulence generate highly structured spatial-temporal dependence [51];

- **Groundwater flow:** hydraulic processes depend on heterogeneous permeability fields [54] and boundary conditions [101];

- **Ocean circulation:** large-scale currents interact with multiscale stochastic forcing [128] and sparse observations [8];

- **Heat diffusion:** spatial propagation depends on physically constrained transport operators [82];

- **Contaminant propagation:** uncertainty evolves dynamically across space and time [31].

Hybrid operator-learning frameworks extend residual hybrid modeling [64] toward broader integrated spatial workflows [20]:

- **Physical constraints:** governing equations [101] and conservation laws provide structural regularization [82];

- **Stochastic uncertainty:** probabilistic representations account for incomplete observations [8] and unresolved variability [31];

- **Observational data:** measurements constrain the latent spatial field [39];

- **Machine learning representations:** neural architectures approximate highly complex nonlinear mappings [52].

This direction represents a transition from purely empirical interpolation toward physically informed spatial inference [101] and operator-based environmental modeling [82].

## 10.6 Spatial validation and predictive realism

A major risk in hybrid modeling is validation bias caused by spatial autocorrelation [106] and spatial sorting bias [96].

Random cross-validation frequently overestimates predictive skill because nearby observations are statistically dependent according to spatial autocorrelation measures [97] and covariance theory [31].

A flexible model may therefore appear highly accurate simply because training and testing samples belong to the same local spatial regime.

Spatially blocked validation provides a more realistic assessment by separating training and testing regions geographically [106], with practical choices of blocking strategy [117] and spatial block implementation [123].

This approach is important because it evaluates whether the model can:

- **Generalize spatially:** prediction quality should remain stable in genuinely unobserved regions [106];

- **Capture transferable structure:** the model should learn environmental relationships [96] rather than memorize local neighborhoods [123];

- **Support operational deployment:** environmental applications often require extrapolation across space [31] and time [8].

This issue is especially important in hybrid models because highly flexible machine learning components may memorize residual spatial patterns without learning physically meaningful structure [106] or calibrated uncertainty [48].

Consequently, spatial validation should be interpreted as a necessary component of defensible spatial inference rather than merely a model selection procedure [106] or a generic random resampling exercise [123].

## 10.7 Why hybrid models are increasingly important

Environmental systems are simultaneously:

- **Nonlinear:** responses often emerge from interacting physical, biological, and chemical processes [52];
- **Multiscale:** different mechanisms operate at local, regional, and continental scales simultaneously [128];
- **Spatially correlated:** nearby observations are statistically dependent through latent spatial continuity [31];
- **Partially observed:** sampling is always finite [130] and often sparse relative to the complexity of the field [8];
- **Physically constrained:** conservation laws [101], transport equations [82], and boundary conditions restrict admissible behavior [39].

No single modeling framework fully captures stochastic spatial dependence [31] and nonlinear representation learning [52] at the same time.

Hybrid methods are therefore increasingly attractive because they combine:

- **Nonlinear predictive flexibility:** machine learning captures complex environmental relationships [77];
- **Probabilistic uncertainty representation:** geostatistics provides interpretable uncertainty linked to spatial structure [54];
- **Spatial dependence modeling:** covariance analysis preserves information about continuity [31] and scale [128];
- **Multiscale integration:** different components of variability can be represented simultaneously [53];

- **Physical interpretability:** hybrid frameworks can incorporate scientific constraints [101] and mechanistic understanding [82].

From a mathematical standpoint, hybridization is not an ad hoc engineering solution. It is a natural consequence of combining different approximation operators acting on complex spatial fields [39] and functional spatial representations [69].

The future of spatial analysis will likely depend increasingly on these integrated frameworks, where stochastic dependence, nonlinear learning, physical modeling, and uncertainty propagation are treated as complementary components of the same inference problem, combining stochastic dependence [31], nonlinear learning [77], and operator-based modeling [82].

# Chapter 11

# Case Study 1: Synthetic Spatial Field with Random Forest, Ordinary Kriging, and Hybrid Regression Kriging

This case study is designed as a controlled spatial experiment [31]. Its purpose is not to reproduce a particular field campaign, but to isolate the relative behavior of three modeling paradigms under a data-generating process where deterministic covariate effects [61] and residual spatial dependence [54] are both present. This setting is useful because it exposes the conditions under which a machine learning model [77], a geostatistical model [31], and a hybrid model [64] succeed or fail.

## 11.1 Synthetic data generation

Let $D \subset \mathbb{R}^2$ denote a bounded spatial domain [31]. We generate $n = 600$ spatial locations

$$\mathbf{s}_i = (x_i, y_i)^\top, \qquad i = 1, \ldots, n, \tag{11.1}$$

sampled uniformly over the domain. The coordinate names $x$ and $y$ are used only to emphasize the two-dimensional geometry of the synthetic field [54]; they play the same role as longitude and latitude in the real-data case study, but without implying a geographic coordinate system.

At each location we define a collection of covariates intended to mimic the structure of the original monthly dataset [67]. The variable names were chosen to preserve the same semantic interpretation as the real application. For example, `elevation` represents a static terrain control [63], `t2m` and `rh2m` represent near-surface temperature and relative humidity [8], `ps` denotes surface pressure, `ws2m` and `wd2m` represent wind speed and direction [51], `allsky_sfc_sw_dwn` is a radiation-type predictor, and `satellite_chirps` and `satellite_merge` are two synthetic satellite rainfall proxies [33]. These names are intentionally retained because they make the synthetic experiment directly comparable to the real-data workflow and keep the interpretation of the model outputs physically meaningful.

The target variable, denoted $Y(\mathbf{s})$, is constructed as

$$Y(\mathbf{s}) = m\Big(\mathbf{z}(\mathbf{s})\Big) + W(\mathbf{s}) + \varepsilon(\mathbf{s}), \tag{11.2}$$

where $m(\cdot)$ is a nonlinear deterministic component [61], $W(\mathbf{s})$ is a spatially correlated Gaussian residual field [31], and $\varepsilon(\mathbf{s})$ is independent measurement noise [37]. In the synthetic design, the mean structure includes nonlinear dependence on elevation, temperature, humidity, wind speed, and the satellite proxies. The residual term $W(\mathbf{s})$ is generated from a covariance model [116] of the form

$$\mathrm{Cov}\Big(W(\mathbf{s}_i), W(\mathbf{s}_j)\Big) = \sigma_W^2 \exp\left(-\frac{\|\mathbf{s}_i - \mathbf{s}_j\|}{a}\right) + \tau^2 \mathbb{I}(i = j), \tag{11.3}$$

where $a$ controls the spatial range [54], $\sigma_W^2$ the sill [31], and $\tau^2$ the nugget effect [130]. This construction ensures that the target field contains a component that is learnable from covariates [67] and another component that is best handled through geostatistical smoothing [54].

To make the case study more realistic, a few additional derived variables are included. For instance, `t2m_range` represents the local diurnal temperature range, `dewpoint_deficit` encodes atmospheric dryness, `satellite_gap` measures disagreement between two synthetic satellite products, and `rh_temp_interaction` introduces a nonlinear coupling between humidity and temperature. These engineered features give the Random Forest a nontrivial covariate space while still leaving a structured spatial residual for the kriging stage to recover.

## 11.2 Spatial partitioning and validation design

To avoid overly optimistic validation caused by spatial autocorrelation [106], the domain is partitioned into spatial blocks using $k$-means clustering on coordinates [123]. The block labels are then used to perform a grouped train-test split, so that entire regions are held out rather than randomly sampled points. This is essential because random pointwise splitting would allow near-duplicate spatial information to appear in both training and testing sets [96], inflating the apparent predictive skill of all methods [106].

In the reported experiment, the dataset contains 600 points, with 458 used for training and 142 for testing. The held-out test set is therefore spatially separated from the training set, giving a more realistic assessment of extrapolation across the domain.

## 11.3 Modeling strategies

Three models are compared.

### 11.3.1 Random Forest

The first benchmark is a Random Forest regression model [11]. It represents the machine learning perspective in which the response is learned from covariates [61] without explicit spatial covariance modeling [31]. Let $\mathbf{x}(\mathbf{s})$ denote the feature vector at location $\mathbf{s}$. The Random Forest estimator can be written abstractly as

$$\widehat{Y}_{RF}(\mathbf{s}) = f_{RF}\Big(\mathbf{x}(\mathbf{s})\Big), \tag{11.4}$$

where $f_{RF}$ is an ensemble of decision trees. In the present case study, the predictor set includes satellite variables, meteorological variables, and engineered nonlinear features such as temperature range, dewpoint deficit, and satellite differences. This model captures nonlinear interactions effectively [11], but it does not explicitly represent spatial dependence [31].

### 11.3.2 Ordinary Kriging

The second benchmark is Ordinary Kriging [92], which uses spatial coordinates and response values through covariance-based prediction [31]. Under ordinary kriging, the random field is decomposed as

$$Y(\mathbf{s}) = \mu + \eta(\mathbf{s}), \tag{11.5}$$

with constant unknown mean $\mu$ and spatially correlated zero-mean residual $\eta(\mathbf{s})$. Prediction at an unsampled location $\mathbf{s}_0$ is obtained through the kriging estimator

$$\widehat{Y}_{OK}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Y(\mathbf{s}_i), \tag{11.6}$$

where the weights $\lambda_i$ are chosen to minimize the prediction variance [31] under unbiasedness constraints [54]. In the implementation, kriging is applied to a log-transformed target to stabilize the marginal distribution [54]:

$$Y^{\star}(\mathbf{s}) = \log\Big(1 + Y(\mathbf{s})\Big). \tag{11.7}$$

Ordinary Kriging is an appropriate reference model because it directly exploits spatial autocorrelation [31]. However, it cannot represent nonlinear dependence on the covariates. In a field with strong deterministic structure, this limitation can be severe.

### 11.3.3 Hybrid regression kriging

The hybrid model combines the strengths of both paradigms [64]. First, a Random Forest estimates the deterministic component:

$$\widehat{m}_{RF}(\mathbf{s}) = f_{RF}\Big(\mathbf{x}(\mathbf{s})\Big). \tag{11.8}$$

Then the residuals [68]

$$r(\mathbf{s}) = Y(\mathbf{s}) - \widehat{m}_{RF}(\mathbf{s}) \tag{11.9}$$

are modeled geostatistically by Ordinary Kriging [54]. The final predictor is

$$\widehat{Y}_{HYB}(\mathbf{s}) = \widehat{m}_{RF}(\mathbf{s}) + \widehat{r}_{OK}(\mathbf{s}). \tag{11.10}$$

A crucial detail is that the kriging stage is fit to out-of-fold residuals rather than in-sample residuals [106]. This avoids leakage and prevents the residual field from being artificially underestimated. The hybrid model therefore becomes a genuine regression kriging method [64] rather than a simple post hoc smoothing layer.

## 11.4 Predictive performance

Table 11.1 summarizes the predictive results on the held-out spatial test set.

**Table 11.1.** Predictive performance for the synthetic spatial case study.

| Model | Test RMSE | Test MAE | Test $R^2$ |
|---|---|---|---|
| Random Forest | 7.9867 | 6.1210 | 0.7363 |
| Ordinary Kriging | 11.8792 | 9.3594 | 0.4166 |
| RF + OK residuals | 7.4535 | 5.8079 | 0.7703 |

The hybrid model achieves the best overall performance, consistent with regression-kriging motivation [64]. Relative to the Random Forest baseline, the hybrid reduces RMSE by approximately 6.7% and improves $R^2$ from 0.7363 to 0.7703. Relative to Ordinary Kriging, the improvement is much larger, with RMSE reduced by approximately 37.3%. These gains are modest in absolute terms but important in a spatial inference setting [31] because they indicate that the residual spatial component contains exploitable structure [54] beyond what the covariates already explain [67].

The comparison also reveals an important methodological point. Ordinary Kriging performs substantially worse than the Random Forest in this synthetic design because the target field is not purely a function of spatial proximity [31]; it is also driven by nonlinear covariate effects [61].

Kriging can interpolate the spatial surface [54], but it cannot reconstruct the covariate-driven mean structure by itself [68]. The Random Forest does better because it captures that deterministic component [11], but it still leaves residual spatial dependence [106]. The hybrid model is strongest because it represents both sources of variation simultaneously.

## 11.5 Observed versus predicted behavior

Figure 11.1 shows the observed-versus-predicted relationships for Ordinary Kriging, Random Forest, and the hybrid model. The kriging scatterplot exhibits systematic shrinkage toward a narrow prediction band, which is expected when the model is driven primarily by spatial averaging [54] and a stationary covariance structure [31]. In this case, Ordinary Kriging uses the spatial configuration of the observations effectively, but it has no direct access to the nonlinear covariate signal used to generate the synthetic field. As a result, high observed values tend to be underpredicted and low observed values tend to be pulled upward, producing a compressed cloud of points around the center of the response range. The Random Forest scatterplot is more dispersed around the 1:1 line and captures the mean response more effectively because it can learn nonlinear relationships [11] between the target variable and environmental predictors [67].

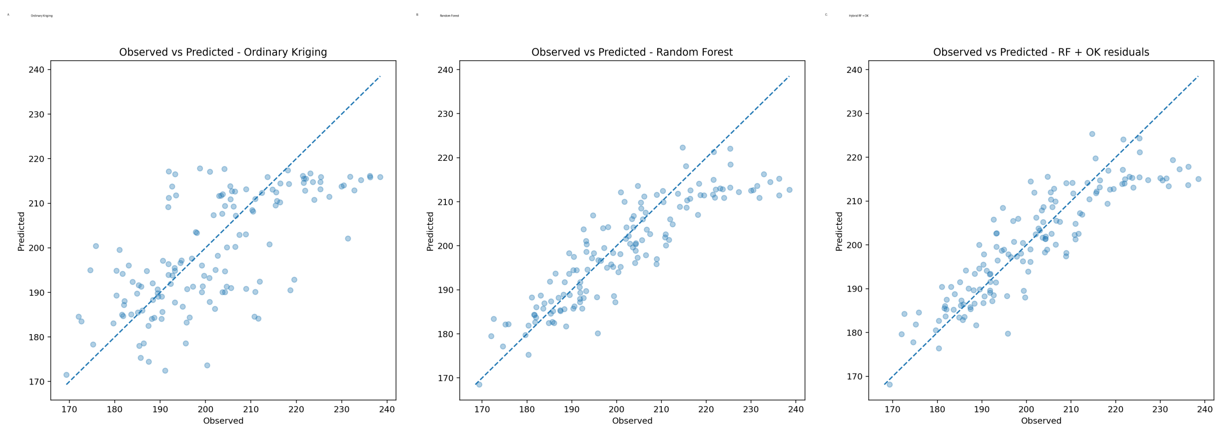


**Figure 11.1. Observed versus predicted values for the synthetic test set. A**, Ordinary Kriging produces predictions that are compressed toward a narrower range, reflecting the smoothing effect of spatial averaging under a stationary covariance model. **B**, Random Forest captures more of the nonlinear covariate-driven signal, so the points follow the identity line more closely, although residual dispersion remains. **C**, The hybrid RF + OK residual model gives the strongest agreement between observations and predictions because it combines nonlinear covariate learning with explicit modeling of residual spatial dependence. Overall, the panel shows why hybrid spatial workflows [64] can outperform either geostatistics [31] or machine learning [77] alone when the target field contains both deterministic environmental structure and spatially correlated residual variation [44].

However, the remaining spread around the identity line indicates that the covariate model alone does not fully remove spatially organized residual variation [106]. The hybrid scatterplot is the closest to the identity line and shows the most balanced dispersion across the prediction range. This pattern is important because it indicates that the hybrid model is not merely improving average error, but also reducing systematic bias across low, medium, and high values. The panel therefore illustrates the practical value of decomposing the spatial prediction problem into two components: a nonlinear mean structure learned from covariates [61] and a residual spatial structure modeled through kriging [64].

## 11.6 Residual structure and spatial diagnostics

The residual map for the hybrid model, shown in Figure 11.2, is useful for assessing whether substantial spatial structure remains after both the machine learning and geostatistical stages. In an ideal hybrid specification, the residuals should appear closer to spatial white noise [31] than in either baseline model.

The presence of localized positive and negative patches indicates that the hybrid still leaves some unresolved fine-scale variation [54], but the amplitude of this structure is reduced relative to a purely covariate-based model [67]. Visually, this map should be read as a diagnostic surface rather than as a final prediction product. If the hybrid model had removed all systematic structure, the residuals would appear as small, randomly scattered deviations with no clear spatial organization. Instead, the remaining clusters reveal where the model still struggles to represent local departures from the broad covariate-driven trend.

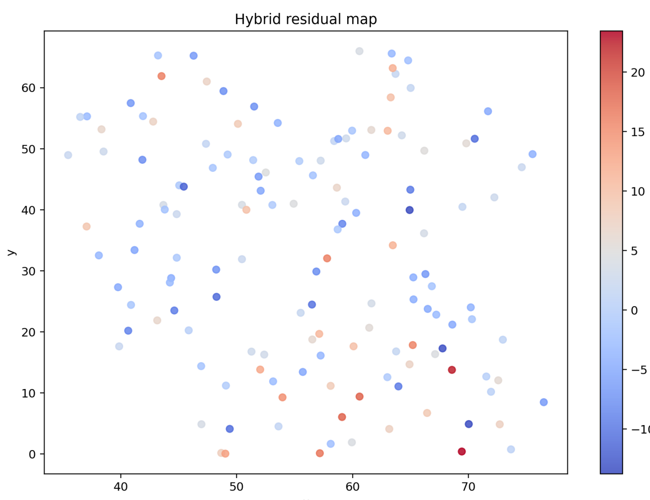


**Figure 11.2. Spatial residuals from the hybrid RF + OK residual model.** Warm and cool colors identify local overprediction and underprediction after combining nonlinear covariate learning with residual kriging.

These zones are important because they may correspond to unmeasured environmental drivers [8], local boundary effects [54], sampling sparsity [130], or short-range spatial processes that are not fully captured by the fitted covariance model [31]. The relatively muted contrast of the map, however, suggests that most of the dominant structure has already been absorbed by the Random Forest mean component and the kriged residual correction. In practical applications, this type of residual visualization is useful for deciding whether additional covariates, a more flexible variogram, local modeling, or targeted new samples are needed.

The permutation importance plot shown in Figure 11.3 indicates that the most influential predictors are the satellite proxy variables [33] and elevation [63].

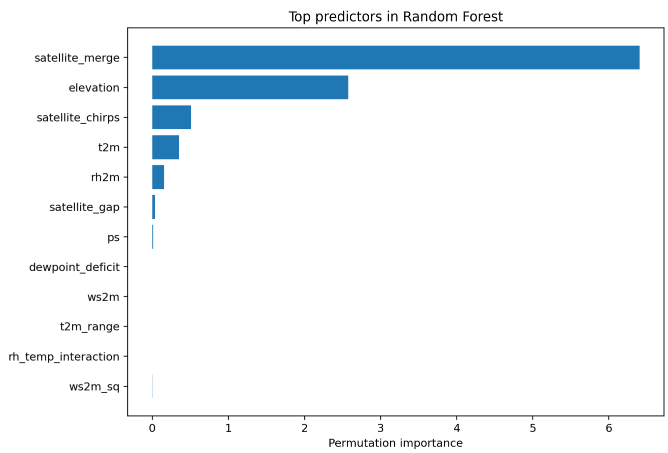


**Figure 11.3. Permutation importance for the Random Forest baseline on the synthetic training/test split.** Higher values indicate covariates that contribute more strongly to predictive accuracy in the fitted Random Forest model.

This is consistent with the data-generating process, which explicitly embeds strong dependence on those terms. Figure 11.3 should therefore be interpreted as a diagnostic check on what the Random Forest has learned from the synthetic experiment. Variables with higher permutation importance are those whose random disruption causes a larger loss of predictive accuracy, indicating that the fitted model relies on them more heavily [11]. The dominance of the satellite proxy variables and elevation is meaningful because these covariates represent the deterministic component of the simulated field. In contrast, predictors with low importance contribute little additional information once the dominant covariates are already included. The importance ranking is therefore not merely descriptive; it confirms that the synthetic experiment is behaving as intended and that the Random Forest is recovering the main nonlinear environmental signal. At the same time, the figure also clarifies why the Random Forest alone is not sufficient: feature importance explains which covariates drive the mean response [11], but it does not account for residual spatial dependence that remains after the covariate effects are learned [31]. This motivates the hybrid strategy, in which the Random Forest captures the deterministic covariate structure [67] and kriging is then used to model the spatially correlated residual component [64].

## 11.7 Interpretation

This case study illustrates the central claim of the chapter in a concrete setting. The synthetic field contains both a nonlinear mean structure [61] and a spatially correlated residual process [31]. A single modeling family is not sufficient to capture both components optimally.

Random Forest is strong at nonlinear regression from heterogeneous covariates [11]. Ordinary Kriging is strong at exploiting spatial continuity [54]. The hybrid model is superior because it decomposes the problem into a deterministic part and a residual spatial part, then models each with the method best suited to it [64]. This is the most important conceptual outcome of the experiment.

The example also clarifies why a hybrid approach can outperform either baseline even when the gain is modest. In spatial analysis, small improvements in RMSE and $R^2$ often correspond to a meaningful reduction in structural error [106], especially when the model is intended for inference [31], uncertainty propagation [54], or decision support rather than only point prediction [8]. The improvement from 7.9867 to 7.4535 in RMSE is therefore methodologically relevant, not just numerically visible.

Finally, Random Forest learns the covariate response [11], Ordinary Kriging learns the spatial covariance [31], and the hybrid learns both [64]. This is precisely the type of decomposition that motivates spatial data fusion [20], residual kriging [64], and hybrid geostatistical machine learning [44] in real applications.

# Chapter 12

# Case Study 2: Factorial Cokriging for Heterogeneous Support

Spatial observations often combine several mechanisms. A soil measurement may contain microscale variability, field-scale management effects, and a broad pedological trend. A satellite embedding may encode local texture, land-cover configuration, and regional climate simultaneously. A seismic attribute may contain a geological signal plus acquisition noise. A single interpolated map mixes all these scales.

Kriging is usually introduced as a best linear unbiased predictor. The same linear machinery can act as a filter when the covariance is decomposed into interpretable pieces. Factorial kriging performs this decomposition for one variable. Factorial *cokriging* performs it for several correlated variables, allowing the scale-specific component of one variable to borrow information from the others.

Three ideas organize the method:

1. The observed random field is an additive combination of latent spatial components.

2. Each component has its own covariance scale, and multivariate coupling at that scale is represented by a coregionalization matrix.

3. The covariance among observations uses all components, while the covariance with the desired target uses only the selected component.

The method therefore answers a question more specific than "what is the value at this location?" It answers "what part of the value is attributable to this mode of spatial variation?"

## 12.1 Random-field setting and stationarity

Let

$$\boldsymbol{Z}(\boldsymbol{x}) = \Big(Z_1(\boldsymbol{x}), \ldots, Z_p(\boldsymbol{x})\Big)^{\mathsf{T}}, \qquad \boldsymbol{x} \in D \subset \mathbb{R}^d, \tag{12.1}$$

be a $p$-variate random field. Under second-order stationarity,

$$\mathbb{E}[\boldsymbol{Z}(\boldsymbol{x})] = \boldsymbol{m}, \tag{12.2}$$

$$\mathrm{Cov}[\boldsymbol{Z}(\boldsymbol{x}), \boldsymbol{Z}(\boldsymbol{x}+\boldsymbol{h})] = \boldsymbol{C}(\boldsymbol{h}), \tag{12.3}$$

where the mean is constant and covariance depends only on separation $\boldsymbol{h}$. Intrinsic stationarity replaces covariance existence with stationary increments, summarized through the matrix-valued semivariogram

$$\boldsymbol{\Gamma}(\boldsymbol{h}) = \frac{1}{2}\mathbb{E}\Big[\{\boldsymbol{Z}(\boldsymbol{x}+\boldsymbol{h}) - \boldsymbol{Z}(\boldsymbol{x})\}\{\boldsymbol{Z}(\boldsymbol{x}+\boldsymbol{h}) - \boldsymbol{Z}(\boldsymbol{x})\}^{\mathsf{T}}\Big]. \tag{12.4}$$

Global stationarity is frequently implausible. Factorial methods are commonly used with a moving neighborhood, so the operational assumption is local stationarity: within each neighborhood, the mean or drift and covariance model are adequate approximations. Neighborhood size then becomes a modeling parameter. A very small neighborhood cannot identify long-range components; a very large neighborhood may violate local homogeneity.

Universal formulations replace a constant mean by

$$\mathbb{E}[Z_r(\boldsymbol{x})] = \sum_{\ell=1}^{q} \beta_{r\ell} f_\ell(\boldsymbol{x}), \tag{12.5}$$

where $f_\ell$ are known drift functions and $\beta_{r\ell}$ are unknown coefficients. Intrinsic random-function theory extends the same logic using generalized covariances and allowable linear combinations [93, 24].

## 12.2 From kriging to filtering

### 12.2.1 Simple kriging

For a centered scalar field observed at $n$ sites, collect data in $\boldsymbol{d} = (Z(\boldsymbol{x}_1), \ldots, Z(\boldsymbol{x}_n))^{\mathsf{T}}$. A linear estimate at $\boldsymbol{x}_0$ is

$$Z^*(\boldsymbol{x}_0) = \boldsymbol{\lambda}^{\mathsf{T}}\boldsymbol{d}. \tag{12.6}$$

Minimizing mean squared error gives

$$\boldsymbol{C}_{DD}\boldsymbol{\lambda} = \boldsymbol{c}_0, \qquad \boldsymbol{\lambda} = \boldsymbol{C}_{DD}^{-1}\boldsymbol{c}_0, \tag{12.7}$$

where $[\boldsymbol{C}_{DD}]_{ij} = C(\boldsymbol{x}_i - \boldsymbol{x}_j)$ and $[\boldsymbol{c}_0]_i = C(\boldsymbol{x}_i - \boldsymbol{x}_0)$. The error variance is

$$\sigma_K^2(\boldsymbol{x}_0) = C(\boldsymbol{0}) - \boldsymbol{c}_0^{\mathsf{T}}\boldsymbol{C}_{DD}^{-1}\boldsymbol{c}_0. \tag{12.8}$$

### 12.2.2 Unknown mean and the block system

With an unknown constant mean, unbiasedness requires $\mathbf{1}^\top \boldsymbol{\lambda} = 1$. The ordinary kriging equations are

$$\begin{bmatrix} \boldsymbol{C}_{DD} & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \mu \end{bmatrix} = \begin{bmatrix} \boldsymbol{c}_0 \\ 1 \end{bmatrix}. \tag{12.9}$$

The extra row and column encode mean reproduction. Universal kriging replaces $\mathbf{1}$ with a design matrix of drift functions. The block formulation is important because factorial kriging changes the right-hand side according to the mean behavior of the requested component.

### 12.2.3 The key filtering step

Suppose

$$Z(\boldsymbol{x}) = T(\boldsymbol{x}) + \sum_{k=1}^{K} Y_k(\boldsymbol{x}), \tag{12.10}$$

where $T$ is a deterministic or slowly varying trend and the $Y_k$ are zero-mean, mutually uncorrelated random fields. Then

$$C_Z(\boldsymbol{h}) = \sum_{k=1}^{K} C_k(\boldsymbol{h}). \tag{12.11}$$

To estimate $Y_k(\boldsymbol{x}_0)$ from observations of the composite field $Z$, use the complete covariance $C_Z$ among the data and the component covariance $C_k$ between the data and target:

$$\boldsymbol{C}_{ZZ} \boldsymbol{\lambda}_k = \boldsymbol{c}_{k0}. \tag{12.12}$$

This is the algebraic heart of factorial kriging. Every scale competes to explain the observed data through $\boldsymbol{C}_{ZZ}$, while only scale $k$ defines what the estimator is trying to recover through $\boldsymbol{c}_{k0}$.

## 12.3 The Linear Model of Coregionalization

### 12.3.1 Nested multivariate structure

For $p$ variables, the LMC writes the covariance and semivariogram as

$$\boldsymbol{C}(\boldsymbol{h}) = \sum_{k=1}^{K} \boldsymbol{B}_k \, \rho_k(\boldsymbol{h}), \tag{12.13}$$

$$\boldsymbol{\Gamma}(\boldsymbol{h}) = \sum_{k=1}^{K} \boldsymbol{B}_k \, g_k(\boldsymbol{h}), \qquad g_k(\boldsymbol{h}) = 1 - \rho_k(\boldsymbol{h}), \tag{12.14}$$

where $\rho_k$ is a standardized correlation model and $\boldsymbol{B}_k \in \mathbb{R}^{p\times p}$ is the coregionalization matrix for component $k$. Its diagonal elements are component variances; its off-diagonal elements are component cross-covariances.

Every $\boldsymbol{B}_k$ must be symmetric positive semidefinite:

$$\boldsymbol{a}^\mathsf{T}\boldsymbol{B}_k\boldsymbol{a} \geq 0 \quad \text{for all } \boldsymbol{a} \in \mathbb{R}^p. \tag{12.15}$$

This condition makes each nested component a valid multivariate covariance. A convenient parameterization is

$$\boldsymbol{B}_k = \boldsymbol{L}_k\boldsymbol{L}_k^\mathsf{T}, \tag{12.16}$$

with lower-triangular $\boldsymbol{L}_k$. Optimizing the entries of $\boldsymbol{L}_k$ makes permissibility structural rather than a post-fit correction.

For an isotropic exponential structure with practical range $a_k$,

$$\rho_k(h) = \exp(-3h/a_k)\,. \tag{12.17}$$

Spherical, Gaussian, Matérn, and anisotropic structures can be used if they are valid covariance functions. A nugget component uses correlation one at zero separation and zero elsewhere.

### 12.3.2 Scale-dependent correlation

The correlation between variables $r$ and $s$ at scale $k$ is

$$\rho_{rs}^{(k)} = \frac{[\boldsymbol{B}_k]_{rs}}{\sqrt{[\boldsymbol{B}_k]_{rr}[\boldsymbol{B}_k]_{ss}}}. \tag{12.18}$$

Two variables can be positively associated at a regional scale and weakly or negatively associated locally. A single Pearson correlation cannot reveal this behavior. Factorial cokriging uses the complete set of $\rho_{rs}^{(k)}$ values through the LMC.

### 12.3.3 Estimating an LMC

For colocated observations, the empirical cross-semivariogram is

$$\widehat{\gamma}_{rs}(\boldsymbol{h}) = \frac{1}{2N(\boldsymbol{h})}\sum_{(i,j)\in\mathcal{P}(\boldsymbol{h})}\{Z_r(\boldsymbol{x}_i) - Z_r(\boldsymbol{x}_j)\}\{Z_s(\boldsymbol{x}_i) - Z_s(\boldsymbol{x}_j)\}, \tag{12.19}$$

where $\mathcal{P}(\boldsymbol{h})$ is a lag bin. A weighted least-squares fit solves

$$\min_{\{\boldsymbol{L}_k, a_k\}} \sum_b \sum_{r\leq s} w_{brs}\left[\widehat{\gamma}_{rs}(h_b) - \sum_{k=1}^{K}[\boldsymbol{L}_k\boldsymbol{L}_k^\mathsf{T}]_{rs}\, g_k(h_b; a_k)\right]^2. \tag{12.20}$$

Weights commonly depend on pair counts and sampling variability. Model selection must consider scientific scale interpretation as well as numerical fit. Nested terms that are nearly indistinguishable in range create unstable factor separation even when the total variogram fits well.

## 12.4 General multivariate factorial cokriging system

### 12.4.1 Observation indexing

Let observation $a$ measure variable $\alpha(a)$ at location $\boldsymbol{x}_a$ and support $V_a$. This notation covers colocated, heterotopic, point, and block data. Stack the $N$ observations in

$$\boldsymbol{d} = (d_1, \dots, d_N)^{\mathsf{T}}. \tag{12.21}$$

For point data, the complete data covariance is

$$[\boldsymbol{C}_{DD}]_{ab} = \sum_{k=1}^{K} [\boldsymbol{B}_k]_{\alpha(a),\alpha(b)} \rho_k(\boldsymbol{x}_a - \boldsymbol{x}_b) + [\boldsymbol{R}]_{ab}, \tag{12.22}$$

where $\boldsymbol{R}$ is optional measurement-error covariance. A nugget may represent microscale process variability, measurement error, or both; these meanings should be separated when replicate information permits.

Consider the component $Y_{k,r}(\boldsymbol{x}_0)$ of target variable $r$. Its data-to-target covariance vector is

$$[\boldsymbol{c}_{k,r}(\boldsymbol{x}_0)]_a = [\boldsymbol{B}_k]_{\alpha(a),r}\, \rho_k(\boldsymbol{x}_a - \boldsymbol{x}_0). \tag{12.23}$$

### 12.4.2 Simple factorial cokriging

When means are known and removed,

$$\boldsymbol{C}_{DD}\boldsymbol{\lambda}_{k,r} = \boldsymbol{c}_{k,r}, \qquad Y^{*}_{k,r}(\boldsymbol{x}_0) = \boldsymbol{\lambda}_{k,r}^{\mathsf{T}}\boldsymbol{d}. \tag{12.24}$$

The left side never changes with the requested factor. After one factorization of $\boldsymbol{C}_{DD}$, many components, target variables, and target locations can be solved efficiently using multiple right-hand sides.

### 12.4.3 Unknown means and drift constraints

Let $\boldsymbol{F} \in \mathbb{R}^{N\times q}$ be the drift design matrix evaluated for each observation and variable. The universal factorial cokriging system is

$$\begin{bmatrix} \boldsymbol{C}_{DD} & \boldsymbol{F} \\ \boldsymbol{F}^{\mathsf{T}} & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda}_{k,r} \\ \boldsymbol{\mu}_{k,r} \end{bmatrix} = \begin{bmatrix} \boldsymbol{c}_{k,r} \\ \boldsymbol{f}_{k,r} \end{bmatrix}. \tag{12.25}$$

For a zero-mean random factor, $\boldsymbol{f}_{k,r} = \boldsymbol{0}$. Thus its weights annihilate every modeled drift:

$$\boldsymbol{F}^{\mathsf{T}}\boldsymbol{\lambda}_{k,r} = \boldsymbol{0}. \tag{12.26}$$

For the complete field, $\boldsymbol{f}_r$ reproduces the target drift. With one unknown constant per variable, $\boldsymbol{F}$ has $p$ columns. Column $s$ is one for observations of variable $s$ and zero otherwise. The complete-field target for variable $r$ uses $\boldsymbol{f}_r = \boldsymbol{e}_r$, whereas a zero-mean factor uses zero.

This distinction explains the familiar signal-plus-noise constraints. If the signal carries the unknown mean and noise has zero mean, signal weights reproduce the mean while noise weights sum to zero. Assigning a unit-sum constraint to every factor would duplicate the mean and destroy coherence.

### 12.4.4 Matrix form for many targets

For $M$ requested target-variable combinations, collect right-hand covariance vectors in $\boldsymbol{C}_{D0}^{(k)}$ and constraints in $\boldsymbol{F}_0^{(k)}$:

$$\begin{bmatrix} \boldsymbol{C}_{DD} & \boldsymbol{F} \\ \boldsymbol{F}^\top & \boldsymbol{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{W}_k \\ \boldsymbol{M}_k \end{bmatrix} = \begin{bmatrix} \boldsymbol{C}_{D0}^{(k)} \\ \boldsymbol{F}_0^{(k)} \end{bmatrix}. \tag{12.27}$$

Predictions are

$$\widehat{\boldsymbol{Y}}_k = \boldsymbol{d}^\top \boldsymbol{W}_k. \tag{12.28}$$

The observation-major ordering

$$(\boldsymbol{x}_1, Z_1), (\boldsymbol{x}_1, Z_2), \ldots, (\boldsymbol{x}_2, Z_1), (\boldsymbol{x}_2, Z_2), \ldots \tag{12.29}$$

is convenient for complete colocated data. Heterotopic data are more naturally stored as explicit location-variable records.

## 12.5 Uncertainty of a factorial estimate

The target component variance is

$$\sigma_{k,r}^2 = [\boldsymbol{B}_k]_{rr} \tag{12.30}$$

for a point target under a unit correlation model. Expanding the mean squared error gives

$$\sigma_{K,k,r}^2 = \sigma_{k,r}^2 - 2\boldsymbol{\lambda}_{k,r}^\top \boldsymbol{c}_{k,r} + \boldsymbol{\lambda}_{k,r}^\top \boldsymbol{C}_{DD} \boldsymbol{\lambda}_{k,r}. \tag{12.31}$$

Using Equation (12.25), this becomes

$$\sigma_{K,k,r}^2 = \sigma_{k,r}^2 - \boldsymbol{\lambda}_{k,r}^\top \boldsymbol{c}_{k,r} - \boldsymbol{f}_{k,r}^\top \boldsymbol{\mu}_{k,r}. \tag{12.32}$$

For a zero-mean factor the final term vanishes. Cross-covariances between factor-estimation errors can also be computed and matter when reconstructed products combine several estimated factors.

The quantity in Equation (12.32) measures uncertainty about the latent component under the fitted model. It does not include LMC parameter uncertainty. Bootstrap variogram fitting, conditional simulation, or Bayesian hierarchical modeling is needed when structural uncertainty is important.

## 12.6 Coherence and additivity

Because the LMC is additive,

$$\boldsymbol{C}_{D0} = \sum_{k=1}^{K} \boldsymbol{C}_{D0}^{(k)}. \tag{12.33}$$

For centered simple cokriging, linearity gives

$$\boldsymbol{W}_{\text{total}} = \boldsymbol{C}_{DD}^{-1}\boldsymbol{C}_{D0} = \sum_{k=1}^{K} \boldsymbol{C}_{DD}^{-1}\boldsymbol{C}_{D0}^{(k)} = \sum_{k=1}^{K} \boldsymbol{W}_{k}. \tag{12.34}$$

Hence the complete prediction equals the sum of component predictions. With an unknown drift, the trend estimator and zero-mean factors must be included in the sum. Their right-hand constraints add to the complete-field constraint. This is the coherence principle emphasized by factorial formulations [89].

Coherence is a powerful software test. For a fixed neighborhood and numerical system, reconstructing the full prediction from all factors plus trend should agree with direct cokriging up to solver tolerance. A failure usually signals inconsistent ordering, omitted structures, duplicated drift constraints, or different support integration across components.

## 12.7 Change of support

### 12.7.1 Areal variables

Let the block average of variable $r$ over support $V$ centered at $\boldsymbol{x}$ be

$$Z_r(V_{\boldsymbol{x}}) = \frac{1}{|V|}\int_{V_{\boldsymbol{x}}} Z_r(\boldsymbol{u})\, d\boldsymbol{u}. \tag{12.35}$$

The covariance between supports $V_{\boldsymbol{x}}$ and $W_{\boldsymbol{y}}$ is

$$C_{rs}^{VW}(\boldsymbol{x},\boldsymbol{y}) = \frac{1}{|V||W|}\int_{V_{\boldsymbol{x}}}\int_{W_{\boldsymbol{y}}} C_{rs}(\boldsymbol{u}-\boldsymbol{v})\, d\boldsymbol{v}\, d\boldsymbol{u}. \tag{12.36}$$

Applying Equation (12.36) to each LMC component yields

$$\boldsymbol{C}_{k}^{VW}(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{B}_k\, \overline{\rho}_{k}^{VW}(\boldsymbol{x},\boldsymbol{y}). \tag{12.37}$$

The same factorial cokriging system then handles point-to-point, point-to-block, and block-to-block prediction by changing covariance integration only.

### 12.7.2 Numerical quadrature

With support discretization points $\boldsymbol{u}_i$, $\boldsymbol{v}_j$ and normalized weights $w_i$, $q_j$,

$$\boldsymbol{C}^{VW}(\boldsymbol{x}, \boldsymbol{y}) \approx \sum_i \sum_j w_i q_j \boldsymbol{C}((\boldsymbol{x} + \boldsymbol{u}_i) - (\boldsymbol{y} + \boldsymbol{v}_j))\,. \tag{12.38}$$

A block has smaller variance than a point because averaging suppresses short-scale variation. For variable $r$, a useful regularization diagnostic is

$$\eta_r(V) = \frac{C_{rr}^{VV}(\boldsymbol{0})}{C_{rr}(\boldsymbol{0})}, \qquad 0 \le \eta_r(V) \le 1. \tag{12.39}$$

Its square root is a standard-deviation correction ratio. Short-range components generally have stronger attenuation than long-range components. This scale-selective attenuation is directly relevant to satellite pixels, field plots, management zones, and depth intervals.

## 12.8 Frequency-domain interpretation

For a stationary process with component spectral densities $f_k(\boldsymbol{\omega})$, the total spectrum is

$$f_Z(\boldsymbol{\omega}) = \sum_k f_k(\boldsymbol{\omega}). \tag{12.40}$$

In an idealized dense and regular setting, the component filter resembles

$$H_k(\boldsymbol{\omega}) = \frac{f_k(\boldsymbol{\omega})}{\sum_j f_j(\boldsymbol{\omega})}. \tag{12.41}$$

Long-range covariance concentrates power at low spatial frequencies; short-range covariance spreads power toward higher frequencies. Factorial kriging is therefore related to Wiener filtering, with the advantage that it works directly with irregular sampling, multivariate dependence, drift constraints, and block supports. Kriging weights act as a spatially adaptive transfer function rather than a fixed convolution kernel.

## 12.9 Filtering, interpolation, and exactness

The complete kriging predictor can honor data exactly when the model and numerical system define an exact interpolator. An individual factorial component generally does not equal the composite observation at a sampled location. This behavior is intentional. A measured value contains every component, while a factorial estimate targets only one latent component.

At sampled locations, factorial cokriging is primarily a probabilistic filter. At unsampled locations, it combines filtering and interpolation. The resulting component map should therefore be assessed by reconstruction, scale behavior, and external scientific meaning rather than by demanding exact reproduction of the composite data.

## 12.10 Worked two-variable model

Consider two variables and two exponential structures. Let

$$\boldsymbol{B}_1 = \begin{bmatrix} 2.0 & 0.5 \\ 0.5 & 1.5 \end{bmatrix}, \quad a_1 = 10, \qquad \boldsymbol{B}_2 = \begin{bmatrix} 4.0 & 2.5 \\ 2.5 & 3.0 \end{bmatrix}, \quad a_2 = 50. \tag{12.42}$$

Both matrices are positive definite. Their scale correlations are

$$\rho_{12}^{(1)} = \frac{0.5}{\sqrt{2.0 \times 1.5}} \approx 0.289, \tag{12.43}$$

$$\rho_{12}^{(2)} = \frac{2.5}{\sqrt{4.0 \times 3.0}} \approx 0.722. \tag{12.44}$$

The variables share much more of their regional variation than their local variation. Variable 2 can therefore contribute strongly to the long-range component of variable 1 even when its local predictive contribution is modest.

The total covariance is

$$\boldsymbol{C}(h) = \boldsymbol{B}_1 \exp(-3h/10) + \boldsymbol{B}_2 \exp(-3h/50). \tag{12.45}$$

To estimate the short-range component of variable 1, build $\boldsymbol{C}_{DD}$ from both terms and build the right-hand vector from the first column of $\boldsymbol{B}_1 \exp(-3h/10)$. For the long-range component, retain the same left-hand matrix and replace the right-hand vector with the first column of $\boldsymbol{B}_2 \exp(-3h/50)$.

## 12.11 Computational workflow

1. **Define variables and supports.** Record coordinate reference system, measurement support, units, timestamps, and missingness. Distances and covariance ranges must use consistent units.

2. **Inspect and transform marginal distributions.** Handle obvious errors and consider transformations when extreme skewness damages linear prediction. Preserve a scientifically defensible back-transformation path.

3. **Calculate direct and cross-semivariograms.** Use common lag bins when variables are colocated. Examine directional variograms before assuming isotropy.

4. **Specify nested scales.** Combine empirical evidence with process knowledge. Nugget, short-range, and regional structures require enough pairs to be separately identifiable.

5. **Fit a permissible LMC.** Enforce positive semidefiniteness of every $\boldsymbol{B}_k$. Inspect eigenvalues, scale correlations, residuals, and sensitivity to starting values.

6. **Regularize covariance to the relevant supports.** Use the same quadrature rules consistently in the data covariance, component right-hand covariance, and target variance.

7. **Choose drift constraints and neighborhood.** Assign drift to the trend or signal component and zero constraints to zero-mean factors. Ensure the local system has enough observations of every relevant variable.

8. **Factorize the common matrix.** Solve linear systems using a symmetric factorization or stable indefinite solver. Avoid explicit matrix inversion.

9. **Solve component right-hand sides.** Batch targets and factors when memory permits. Return weights, multipliers, variances, and condition diagnostics with predictions.

10. **Validate scientifically and numerically.** Check additivity, cross-validation, residual spatial structure, neighborhood sensitivity, support convergence, and stability under plausible LMC alternatives.

## 12.12 Reference implementation pattern

The following GeoSupportFM-style example mirrors the mathematical workflow:

```
import numpy as np
from geosupportfm.coregionalization import (
    LMCStructure,
    LinearModelOfCoregionalization,
    empirical_variogram_matrix,
    fit_lmc,
)
from geosupportfm.kriging import factorial_cokriging, ordinary_cokriging
from geosupportfm.support import SpatialSupport, regularize_lmc

# 1. Experimental direct and cross-semivariograms.
lags, gamma, pair_counts = empirical_variogram_matrix(
```

```
    sample_coords,
    sample_values,
    n_lags=12,
    pair_min=5,
)

# 2. Initial nested model and constrained LMC fit.
initial = (
    LMCStructure("exponential", 10.0, np.eye(2)),
    LMCStructure("exponential", 50.0, np.eye(2)),
)
fit = fit_lmc(
    lags,
    gamma,
    initial,
    weights=pair_counts,
    fit_ranges=True,
)
lmc = fit.lmc

# 3. A 10 m by 10 m target support using midpoint quadrature.
pixel = SpatialSupport.rectangle((10.0, 10.0), discretization=5)
support_profile = regularize_lmc(
    lmc,
    pixel,
    lags=np.linspace(0.0, 100.0, 21),
)

# 4. Extract short and long components at block support.
short = factorial_cokriging(
    sample_coords,
    sample_values,
    target_coords,
    lmc,
    component=0,
    target_support=pixel,
)
long = factorial_cokriging(
    sample_coords,
    sample_values,
    target_coords,
    lmc,
    component=1,
    target_support=pixel,
)
```

```
# 5. Complete ordinary block-cokriging prediction.
total = ordinary_cokriging(
    sample_coords,
    sample_values,
    target_coords,
    lmc,
    target_support=pixel,
)
```

When the factors are modeled as zero mean and the complete field contains an unknown mean, the component sum alone need not equal the ordinary prediction; the estimated mean or trend completes the decomposition. For centered data with a known mean, simple factorial components should sum directly to the simple cokriging prediction.

## 12.13 Foundation-model embeddings

Let $\boldsymbol{E}(\boldsymbol{x}) \in \mathbb{R}^p$ denote a geospatial embedding. Factorial cokriging can identify latent axes or low-dimensional projections that share local or regional structure with an environmental target $Y(\boldsymbol{x})$. A practical LMC might include $Y$, selected embedding dimensions, and physical covariates.

Three aspects require care:

1. **Dimensionality.** A full LMC for dozens of embedding dimensions has many parameters. Variogram-aware screening, principal components, or scientifically defined projections should precede fitting.

2. **Geometry.** Unit-normalized embeddings lie on a hypersphere. Euclidean component cokriging is a local linear approximation. Aggregated embeddings may need renormalization when cosine geometry is required downstream.

3. **Support.** A point field observation and a satellite pixel do not share the same support. Equation (12.38) aligns their covariances before scale decomposition.

The resulting factors can distinguish local embedding anomalies from regional representation gradients. They can also provide support-aware block features for a downstream scientific machine-learning model.

## 12.14 Diagnostics and common failure modes

Additional checks include:

- plot every direct and cross-semivariogram with its fitted LMC;

| Symptom | Likely cause | Action |
|---|---|---|
| Negative covariance eigenvalues | Impermissible LMC matrices | Fit $\boldsymbol{B}_k = \boldsymbol{L}_k \boldsymbol{L}_k^\top$ and inspect numerical tolerance. |
| Unstable component maps | Similar nested ranges or weak lag information | Reduce model complexity, collect more pairs, or constrain ranges. |
| Large alternating weights | Ill-conditioned covariance or redundant variables | Scale variables, remove redundancy, use local neighborhoods, and inspect the condition number. |
| Factors do not reconstruct total | Inconsistent drift constraints, support, or ordering | Verify covariance additivity and right-hand-side constraints. |
| Block variance exceeds point variance | Incorrect normalization or quadrature weights | Normalize support weights and test increasing discretization. |
| Smooth component interpreted as truth | Structural model uncertainty ignored | Compare plausible LMCs and perform conditional simulation or bootstrap. |
| Cross-validation looks good but factors lack meaning | Nested decomposition chosen only by fit | Use process knowledge and external scale-specific validation. |

**Table 12.1.** Typical factorial cokriging diagnostics.

- inspect eigenvalues and scale correlations for every $\boldsymbol{B}_k$;
- repeat the analysis across neighborhood sizes and anisotropy models;
- test quadrature convergence for every support geometry;
- map component uncertainty beside component estimates;
- compare reconstructed and direct complete-field predictions;
- evaluate whether removed "noise" is correlated with external variables, which may reveal a mislabeled scientific signal.

## 12.15 Interpretive limits

Factorial components are latent conditional expectations defined by the fitted covariance decomposition. They are not directly observed physical processes. Calling a short-range term "noise" or a long-range term "trend" is a scientific interpretation requiring contextual evidence. The same empirical variogram may admit several nested decompositions, and those decompositions can produce different factor maps while giving similar total predictions.

Factorial inference is also an ecological inference: a component is inferred from observations of the composite process. This can create misleading associations when scale definitions are poorly justified. The method is most credible when spatial scales correspond to known mechanisms, sampling density resolves those scales, and external information supports the interpretation.

# Chapter 13

# Concluding remarks

Spatial modeling should not be reduced to interpolation [31]. At its core, it is the problem of inferring spatially structured functions [110] from incomplete [8], noisy [37], and multiscale observations [128]. Once the problem is framed this way, the apparent divide between geostatistics and machine learning becomes less important than the mathematical structure they share.

Geostatistics provides a rigorous probabilistic language for:

- **Spatial dependence:** covariance models [31] and variogram models [54] describe how similarity changes with distance and direction.
- **Uncertainty propagation:** kriging variance [31], conditional simulation [54], and probabilistic thresholds provide interpretable measures of confidence.
- **Multiscale stochastic variability:** nested structures [128] and coregionalization [53] allow the decomposition of spatial variation across different scales.
- **Support-aware inference:** geostatistical methods can explicitly account for the fact that point values [31], pixels [33], and block averages [57] are not interchangeable.

Machine learning provides a complementary language for:

- **Nonlinear approximation:** flexible models can capture complex relationships [61] that are difficult to express with fixed covariance forms [77].
- **High-dimensional covariates:** remote sensing [33], topography [63], climate [8], and ancillary data can be exploited jointly [67].
- **Representation learning:** deep architectures can discover nonlinear features [86] from structured spatial inputs [52].

- **Large-scale prediction:** modern ML systems can handle very large datasets [52] more efficiently than classical kriging in many settings [116].

The opposition between these two traditions is therefore mostly pedagogical. From a functional analytic perspective, both are approximation methods on structured function spaces. Kriging is a projection problem [92]. Kernel methods are RKHS approximations [113]. Regularized inverse problems solve constrained reconstruction tasks [39]. Neural operators learn mappings between function spaces [82]. These are different expressions of a shared mathematical theme.

This viewpoint has several consequences. First, it clarifies why geostatistics and machine learning can be combined effectively rather than used as competing dogmas. Second, it explains why uncertainty, scale, and support matter so much in spatial analysis. Third, it shows that the most useful model is often not the one that is conceptually pure, but the one that best matches the structure of the data and the decision problem.

In practical terms, the strengths and limitations of each framework are complementary:

- **Geostatistics is strongest when spatial correlation dominates:** it provides principled interpolation [54], interpretable covariance structure [31], and explicit uncertainty estimates [116].

- **Machine learning is strongest when nonlinear predictors dominate:** it handles complex interactions [61], heterogeneous inputs [67], and high-dimensional feature spaces [52].

- **Hybrid methods are strongest when both effects matter:** they combine the explanatory power of covariates [68] with the residual structure captured by spatial dependence [64].

Hybrid models are increasingly important because environmental systems are simultaneously nonlinear [52], spatially correlated [31], partially observed [8], multiscale [128], and physically constrained [101]. No single framework fully captures all of these properties. A purely geostatistical model may underuse rich covariates [68]. A purely machine learning model may ignore residual dependence [106] and produce weak uncertainty estimates [48]. A hybrid model can often do both jobs more credibly [44].

The broader conclusion is that spatial analysis should be treated as functional inference over complex fields [69] rather than as a choice between interpolation algorithms [31]. This shift in perspective encourages models that are more faithful to the data-generating process [8] and more useful for decision-making [54]. It also suggests a direction for future work: methods that integrate stochastic dependence [31], nonlinear approximation [77], physical constraints [101], support effects [57], and uncertainty propagation [54] within a single coherent framework.

In that sense, the field is moving from map construction toward mathematically informed spatial inference [82].

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